

Kinesin Autoinhibition Requires Elbow Phosphorylation

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In their **important** manuscript, Chen et al. investigate the phospho-regulation of the *C. elegans* kinesin-2 motor protein OSM-3, revealing that the kinase, NEKL-3, phosphorylates a serine/threonine patch at the hinge region of the motor to mediate autoinhibition until it reaches the ciliary middle segment. The findings are supported by robust genetic data, in vivo imaging, and motility assays with wild-type and mutant motors. Overall, the study provides a **compelling** contribution to understanding the regulation of OSM-3 kinesin activity both on the molecular and cellular levels.

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Abstract

Kinesin motor proteins facilitate microtubule-based transport by converting chemical energy into mechanic forces, but this activity is autoinhibited until cargo is loaded. Regulatory mechanisms underlying this autoinhibitory conformation are not well understood. Here, we show that a NEver in mitosis Kinase NEKL-3 directly phosphorylates a flexible elbow region between two coiled coil domains connecting the motor head and tail of an intraflagellar transport kinesin, OSM-3. The phosphor dead (PD) mutation, but not phosphor mimic (PM) mutation, induce constitutive motility of OSM-3 in vitro. Using knock-in animals, we discovered that both PD and PM mutations shorten the *C. elegans* sensory cilia. The constitutively active OSM-3PD fails to enter cilia and abnormally accumulates in neurites, mimicking another hyperactive mutation, OSM-3G444E. Conversely, OSM-3PM enters cilia but moves at a reduced speed, indicating an inhibitory role of elbow phosphorylation in kinesin motility. These findings highlight the crucial role of elbow phosphorylation in regulating kinesin autoinhibition.

Introduction

Kinesin motor proteins, integral to intracellular transport along microtubules, are tightly regulated for precise cargo delivery within cells (Burute & Kapitein, 2019 [DOI](#); Christensen & Reck-Peterson, 2022 [DOI](#); Hirokawa et al., 2009 [DOI](#); G. Ou & J. M. Scholey, 2022 [DOI](#)). Dysregulation of kinesin activity is linked to neurodegenerative disorders like Alzheimer's and Parkinson's diseases (Brady & Morfini, 2017 [DOI](#); Sleight et al., 2019 [DOI](#)). Understanding these regulatory mechanisms is vital for elucidating disease mechanisms and developing potential therapeutic strategies. Mechanisms controlling kinesin activation involve protein interactions, post-translational modifications, and cargo binding (Cason & Holzbaur, 2022 [DOI](#); Verhey & Hammond, 2009 [DOI](#)). Protein phosphorylation is recognized for its regulatory role in a variety of proteins, yet its impacts on kinesin motility are not fully understood (Banerjee et al., 2021 [DOI](#); DeBerg et al., 2013 [DOI](#); Espeut et al., 2008 [DOI](#); Liang et al., 2014 [DOI](#); Sato-Yoshitake et al., 1992 [DOI](#); Schäfer et al., 2008 [DOI](#)).

Cilia serve as a unique model for studying how microtubule-based motor proteins are regulated in specific cellular regions (Anvarian et al., 2019 [DOI](#); Nachury & Mick, 2019 [DOI](#)). The formation of cilia relies on bidirectional intraflagellar transport (IFT) along microtubules within the axoneme (Ishikawa & Marshall, 2011 [DOI](#); Klena & Pigino, 2022 [DOI](#); Taschner & Lorentzen, 2016 [DOI](#)). At the ciliary base, kinesin-2 family proteins undergo conformational changes to transport IFT-particles loaded with ciliary precursors to the axonemal tip (G. Ou & J. M. Scholey, 2022 [DOI](#); Taschner & Lorentzen, 2016 [DOI](#)). After unloading cargo, kinesin-2 is deactivated, while dynein-2 is activated to recycle the anterograde IFT machinery (G. Ou & J. M. Scholey, 2022 [DOI](#); Prevo et al., 2017 [DOI](#); Taschner & Lorentzen, 2016 [DOI](#)). In *Caenorhabditis elegans*, kinesin-II and OSM-3 collaborate to construct sensory cilia in chemosensory neurons (Ou et al., 2005 [DOI](#); Guangshuo Ou & Jonathan M. Scholey, 2022; Prevo et al., 2015 [DOI](#)). Initially, kinesin-II transports IFT-particles, building the middle ciliary segments, while OSM-3 is inactive and transported by kinesin-II (Mitra et al., 2024 [DOI](#); Ou et al., 2005 [DOI](#); Guangshuo Ou & Jonathan M. Scholey, 2022; Prevo et al., 2015 [DOI](#)). Later, kinesin-II transfers IFT particles to OSM-3, activating it to convey cargo molecules and assemble the distal axoneme (Mitra et al., 2024 [DOI](#); Ou et al., 2005 [DOI](#); Guangshuo Ou & Jonathan M. Scholey, 2022; Prevo et al., 2015 [DOI](#)). The molecular mechanisms that govern the activation and inactivation of kinesin-II and dynein-2 along the cilia are still unclear. Additionally, the regulation of OSM-3's regional motility presents an even greater mystery: Deleting OSM-3 only disrupts the distal axoneme without affecting the ciliary middle region (Mitra et al., 2024 [DOI](#); Ou et al., 2005 [DOI](#); Guangshuo Ou & Jonathan M. Scholey, 2022; Prevo et al., 2015 [DOI](#)). It remains unclear how OSM-3 becomes enriched in the distal domain than the middle segment.

The “elbow” of a kinesin refers to a flexible region situated between two coiled-coil domains, linking the motor head and tail of the kinesin protein (Tan et al., 2023 [DOI](#); Weijman et al., 2022 [DOI](#)). This junction imparts necessary flexibility and facilitates interactions between the globular motor head and the rigid stalk-like tail (Tan et al., 2023 [DOI](#); Verhey & Hammond, 2009 [DOI](#); Weijman et al., 2022 [DOI](#)). The elbow region is thus considered to serve as a pivotal regulatory center, governing the motor's activity and facilitating precise spatiotemporal control of intracellular transport processes (Tan et al., 2023 [DOI](#); Weijman et al., 2022 [DOI](#)). However, the post-translational modification of the kinesin elbow and its physiological consequences have not been elucidated. In our previous study, a genetic screen targeting the OSM-3(G444E) hyperactive mutation identified NEKL-4, a member of the NIMA kinase family, as a suppressor of this phenotype (Peishan Yi et al., 2018 [DOI](#)). This finding, combined with reports that NIMA kinases regulate ciliary processes independently of their canonical mitotic roles (Chivukula et al., 2020 [DOI](#); Fry et al., 2012 [DOI](#); Smith et al., 2006 [DOI](#); Thiel et al., 2011 [DOI](#)), prompted us to investigate whether NIMA kinases modulate OSM-3-driven intraflagellar transport. We hypothesized that NEKL-3/4, as paralogs within this family, might directly phosphorylate OSM-3 to regulate its motility. In this study, we demonstrate that the NEKL-3 kinase directly phosphorylates the elbow of OSM-3 kinesin. Our in vitro and in vivo findings suggest that

elbow phosphorylation exerts an inhibitory effect on OSM-3 motility. We show that elbow phosphorylation inhibits OSM-3 motility in the soma and dendrites of sensory neurons, while its dephosphorylation is necessary for activating OSM-3 in sensory cilia.

Results

NEKL-3 phosphorylates the OSM-3 kinesin's elbow

To determine whether NIMA kinase family members could directly phosphorylate OSM-3, we purified prokaryotic recombinant *C. elegans* NEKL-3/NEKL-4 and OSM-3 protein in order to perform in vitro phosphorylation assays. We were able to obtain active recombinant NEKL-3 but not NEKL-4. The in vitro phosphorylation assays showed that NEKL-3, directly phosphorylates OSM-3 (Fig. 1A-B, Appendix Table S1). Subsequent mass spectrometric analysis revealed phosphorylation at residues 487-490, which localize to the conserved “YSTT” motif within OSM-3's C-terminal tail region. (Fig. 1A-E, Fig. S1, Appendix Table S1). To gain structural insights from this motif, we employed LocalColabFold based on AlphaFold2 to predict the dimeric structure of OSM-3 (Evans et al., 2022; Jumper et al., 2021; Mirdita et al., 2022). The highest-confidence model was selected for further analysis (Fig. 1C, Fig. S2). These four phosphorylated “YSTT” residues are situated at the end of the hinge2 region, linking coiled-coil1 and coiled-coil2 (Fig. 1A, C-D). In kinesin-1, this inter-coiled-coil junction is known as the “elbow”, essential for adopting a compact conformation for autoinhibition (Tan et al., 2023; Weijman et al., 2022). Accordingly, we have designated the region comprising amino acids 487-490 as the “elbow” of OSM-3. Although the elbow region is prevalent throughout the kinesin superfamily, the molecular regulation of the elbow conformation remains to be illustrated.

Phosphorylation at the elbow inhibits the motility of the OSM-3 kinesin in *C. elegans*

To investigate the in vivo effects of phosphorylation and dephosphorylation at the “elbow” region of OSM-3, we utilized genome-editing techniques to generate knock-in worms harboring phospho-dead (PD) and phospho-mimic (PM) mutations. Specifically, we replaced the amino acids 487-490 YSTT with FAAA for the PD mutation and DDEE for the PM mutation in the *C. elegans* OSM-3::GFP genome (Fig. 1D-E). The wild-type (WT) OSM-3::GFP facilitates anterograde IFT to construct the distal ciliary segments of sensory neurons, and OSM-3::GFP fluorescence localizes along the ciliary distal segment (Fig. 2A-B). However, OSM-3PD::GFP was absent from cilia and excluded from cell bodies, instead forming bright puncta around the axons of sensory neurons (Fig. 2A-B). By introducing an IFT marker IFT70/DYF-1::mScarlet into *osm-3pd* worms, we revealed a marked reduction in ciliary length to $2.15 \pm 0.37 \mu\text{m}$ compared to WT animals of $7.74 \pm 0.94 \mu\text{m}$, consistent with the absence of OSM-3 kinesin within the cilia (Fig. 2B-C). The shortened cilia length and the formation of abnormal puncta at the neurite tip resemble the phenotype observed in our previously characterized OSM-3G444E mutation, which disrupts autoinhibition and leads to hyperactivation of OSM-3 (Xie et al., 2024). Conversely, OSM-3PM::GFP fluorescence did not form puncta but localized within the sensory cilia, albeit with shortened cilia of $4.89 \pm 1.14 \mu\text{m}$ and reduced IFT speed at both middle and distal segment, indicating impaired OSM-3 motility (Fig. 2B-D, Fig. S3).

These findings indicate that OSM-3-PM is in an autoinhibited state capable of ciliary delivery, yet fails to achieve full activation due to defective dephosphorylation. This incomplete activation results in suboptimal motor function and intermediate ciliary length phenotypes (Fig. 2B-C). In contrast, OSM-3-PD exhibits constitutive activation leading to aggregation into axonal puncta, which completely abolishes its ciliary entry capacity (Fig. 2A-B). Our findings collectively suggest that NEKL-3-mediated phosphorylation of the OSM-3 elbow domain serves as a critical

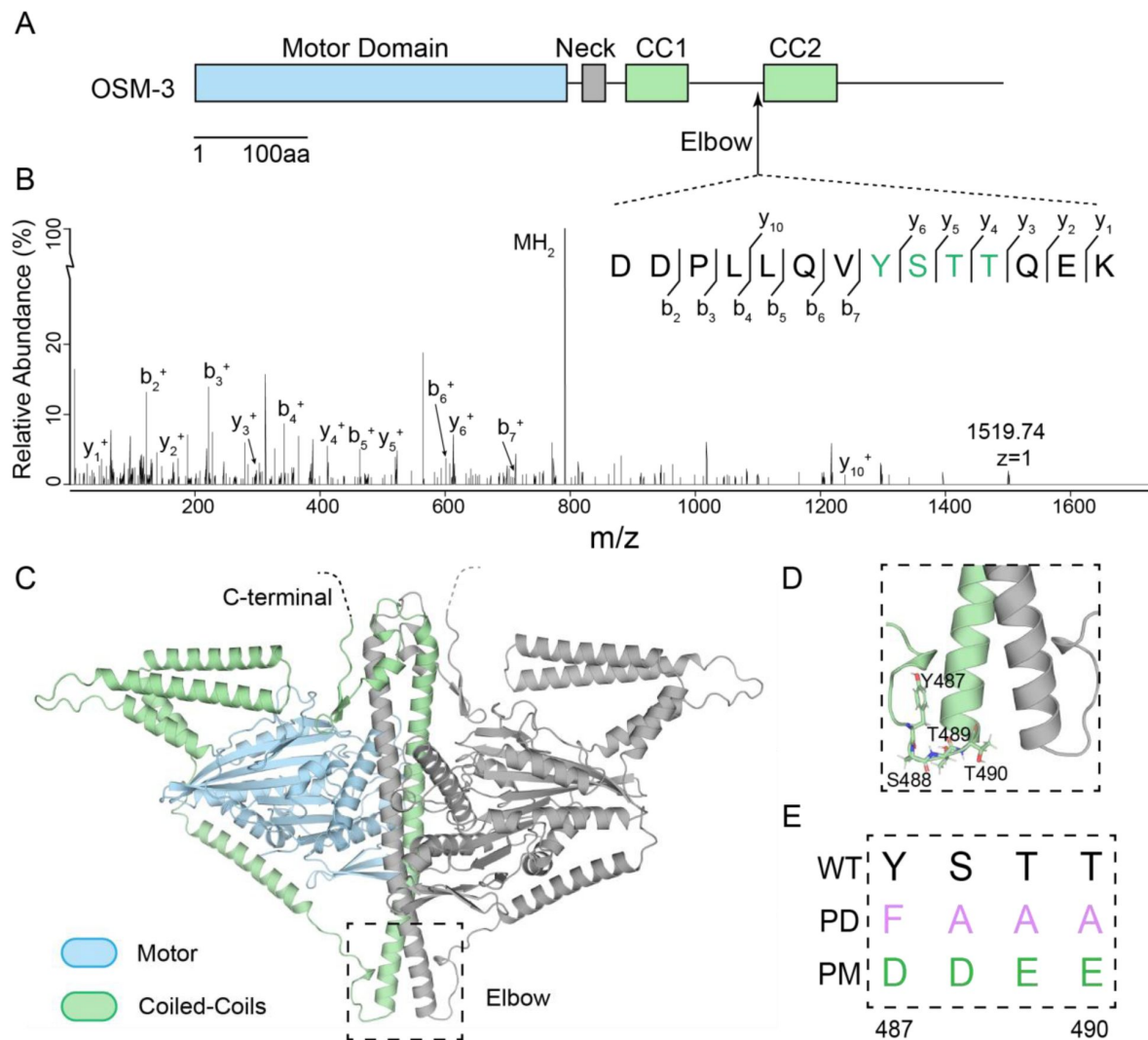


Fig. 1.

NEKL-3 phosphorylates OSM-3 at its “elbow”

(A) Schematic of the full-length OSM-3. Motor domain (blue), neck (gray) and coiled-coils (green) are indicated. CC, coiled-coil. (B) Mass spectrum of an OSM-3 peptide that was phosphorylated by NEKL-3. Phosphorylated gel bands were subjected to MS analysis searching for phosphorylation modifications. Residues 487 to 490 of OSM-3 were phosphorylated and were marked by green color. (C) Phosphorylated 488-490 residues are at the “elbow” of OSM-3. It shows the overall structure of the homodimeric OSM-3 predicted by alphafold2. The dashed square marks the “elbow” region and is zoomed-in in (D). (E) Genome editing constructs of the elbow, showing the PD (phosphor-dead) and PM (phosphor-mimic) sequences comparing to wild type. Residue 487-490 of OSM-3 was edited to “FAAA” for PD strain or edited to “DDEE” for PM strain. Abbreviations: Y, Tyr; S, Ser; T, Thr; F, Phe; A, Ala; D, Asp; E, Glu.

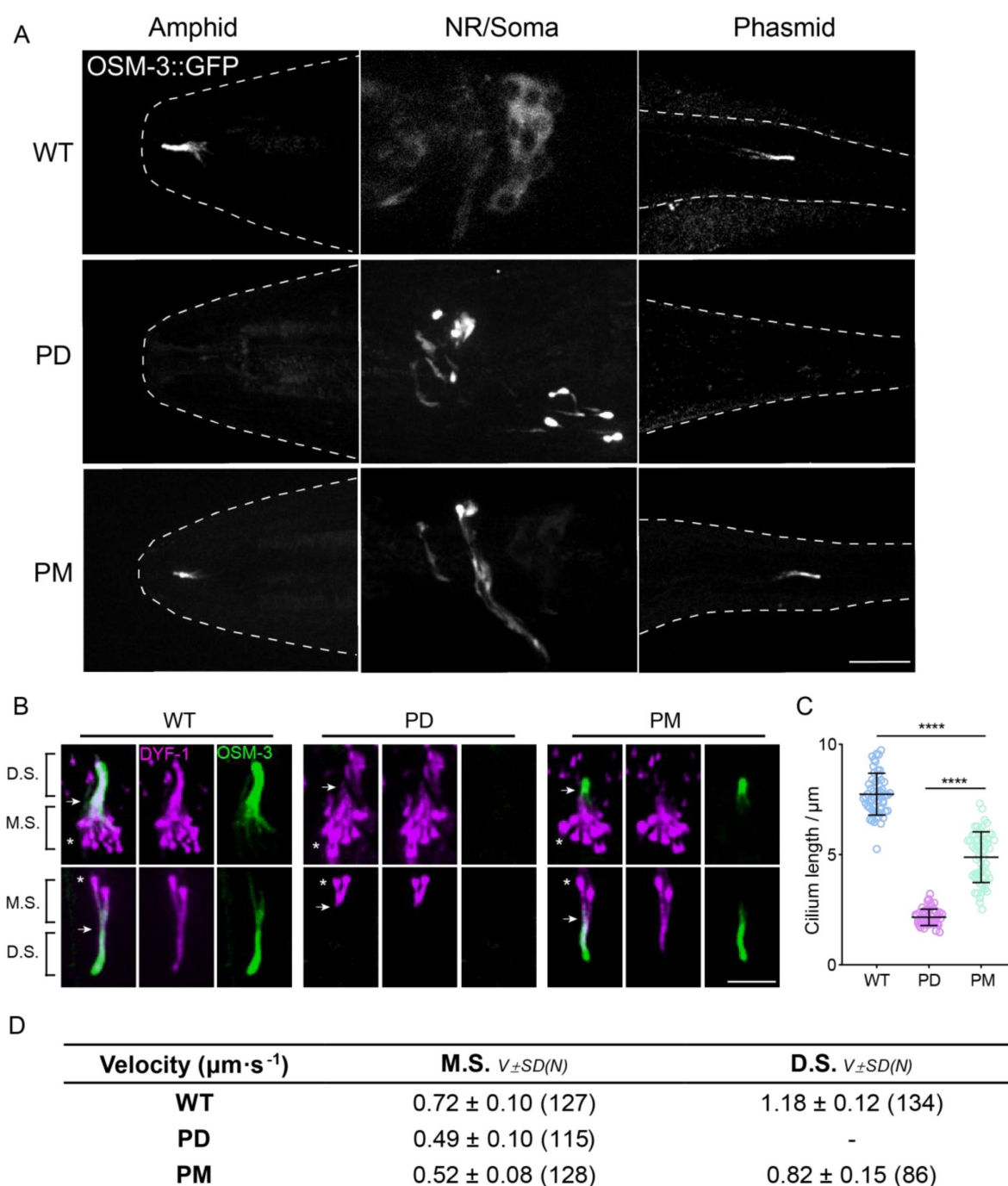


Fig. 2.

Phosphorylation at the elbow of OSM-3 is inhibitive in vivo

(A) Representative images of the phospho-dead (PD) and phospho-mimic (PM) knock-in worms showing their OSM-3 signal at amphid cilia, amphid neuronal soma and phasmid cilia, respectively. The contours of the worms are marked by white dashed lines. Scale bar, 10 μm . (B) Representative images of the cilia from PD and PM worms marked by the ciliary marker DYF-1::mScarlet. White arrows indicate the junction between middle and distal segments while the asterisks indicate the ciliary base. M.S., middle segment; D.S., distal segment. Scale bar, 5 μm . (C) Statistics of the cilium length of the strains showed in (B). The lengths of DYF-1::mScarlet signals were measured and analyzed. (D) IFT velocities of PD and PM worms. ****, $p < 0.0001$, analyzed by one way ANOVA, p values were adjusted by BH method.

regulatory mechanism suppressing motor activity. Under physiological conditions, this phosphorylation-dependent autoinhibition likely prevents precocious activation during cytoplasmic transport, ensuring proper delivery of motor complexes to the cilium. Subsequent dephosphorylation at the ciliary base then activates OSM-3, enabling its processive movement required for intraflagellar transport.

While *nekl-3* null mutants are inviable due to essential mitotic roles (Barstead et al., 2012 [↗](#)), conditional knockout (cKO) of *nekl-3* in ciliated neurons (Huang et al., JCB, 2025, in press, 10.1083/jcb.202407152) revealed its critical role in regulating OSM-3 dynamics. In *nekl-3* cKO animals, OSM-3—normally enriched in the ciliary distal segment—redistributed uniformly along the cilium, concomitant with premature activation of anterograde motility in the middle ciliary region (Huang et al., JCB, 2025, in press, 10.1083/jcb.202407152). This phenotype aligns with our model wherein NEKL-3 phosphorylation suppresses OSM-3 activity, ensuring spatiotemporal regulation of IFT.

Structural models of the OSM-3 kinesin and its mutants

To explore the impact of the elbow phosphorylation on the OSM-3 kinesin, we first predicted a model of the OSM-3 monomer using LocalColabFold and relaxed the mutated models with PyRosetta to calculate the energy-minimized conformations of these mutations (**Fig. 3A** [↗](#)) (Chaudhury et al., 2010 [↗](#); Evans et al., 2022 [↗](#); Jumper et al., 2021 [↗](#); Mirdita et al., 2022 [↗](#)). The WT and PM models resulted in similar conformations, with their tails binding to the motor head and forming a reverse β -sheet secondary structure (**Fig. 3A** [↗](#)). The motor-tail interaction, reported in other motors, typically results in an autoinhibitory state (Coy et al., 1999 [↗](#); Dietrich et al., 2008 [↗](#); Espeut et al., 2008 [↗](#); Friedman & Vale, 1999 [↗](#)). In contrast, the G444E mutation, a characterized hyperactive mutation, showed a head-tail dissociation conformation (**Fig. 3A** [↗](#)) (Imanishi et al., 2006 [↗](#); Xie et al., 2024 [↗](#)). Similarly, the OSM-3PD protein also exhibited a head-tail dissociation conformational change during relaxation (**Fig. 3A** [↗](#)).

To trace the clues of the conformational changes, we evaluated the energy of pre-relaxed structures around the phosphorylated residues (**Fig. 3B-C** [↗](#), **Fig. S4** [↗](#), Appendix Table S2). Compared to the PM model of OSM-3, the PD-mutated side chains had lower side-chain rotamer energy (The ‘fa_dun’ term) (Alford et al., 2017 [↗](#)), likely due to the larger and higher charged residues in the PM state. However, the PD mutation residues exhibited higher folding/unfolding free energy (The ‘ref’ term) (Alford et al., 2017 [↗](#)), indicating an unstable folding state of the mutated loop (**Fig. 3B-C** [↗](#), **Fig. S4** [↗](#), Appendix Table S2). This instability may explain the observed conformational change leading to the autoinhibition release of OSM-3PD.

Biochemical analyses reveal the role of elbow phosphorylation in OSM-3 motility

To directly observe the effects of phosphor mutants in the elbow region of OSM-3 in microtubule binding and motility, we examined the microtubule-stimulated ATPase activity, microtubule gliding activity and single molecular movement of the WT and mutant OSM-3 variants using prokaryotic recombinant proteins (**Fig. 4A-F** [↗](#), **Fig. S5** [↗](#)). The WT full-length OSM-3 exhibited minimal ATPase activity (6.97% of KHC) due to its autoinhibition (**Fig. 4A** [↗](#)). As previously reported, the hyperactive OSM-3G444E mutation showed a significant increase in ATPase activity (131.65% of KHC) (**Fig. 4A** [↗](#)) (Imanishi et al., 2006 [↗](#); Xie et al., 2024 [↗](#)). Consistent with our structural predictions, the OSM-3PD mutation also led to an upregulation of ATPase activity (98.15% of KHC) to a level similar to OSM-3G444E (**Fig. 4A** [↗](#)). In contrast, OSM-3PM caused a slight increase in ATPase activity (34.28% of KHC) compared to WT but was markedly lower than OSM-3G444E or OSM-3PD (**Fig. 4A** [↗](#)). The PM mutant’s partial ATPase activity (34.28% of KHC) might arise from imperfect phosphomimicry—while the DDEE substitution introduces negative charges, it lacks the steric bulk of phosphorylated tyrosine (pY487). And this incomplete mimicry allows residual autoinhibition, sufficient to limit ciliary construction in vivo. Notably, the open-state OSM-

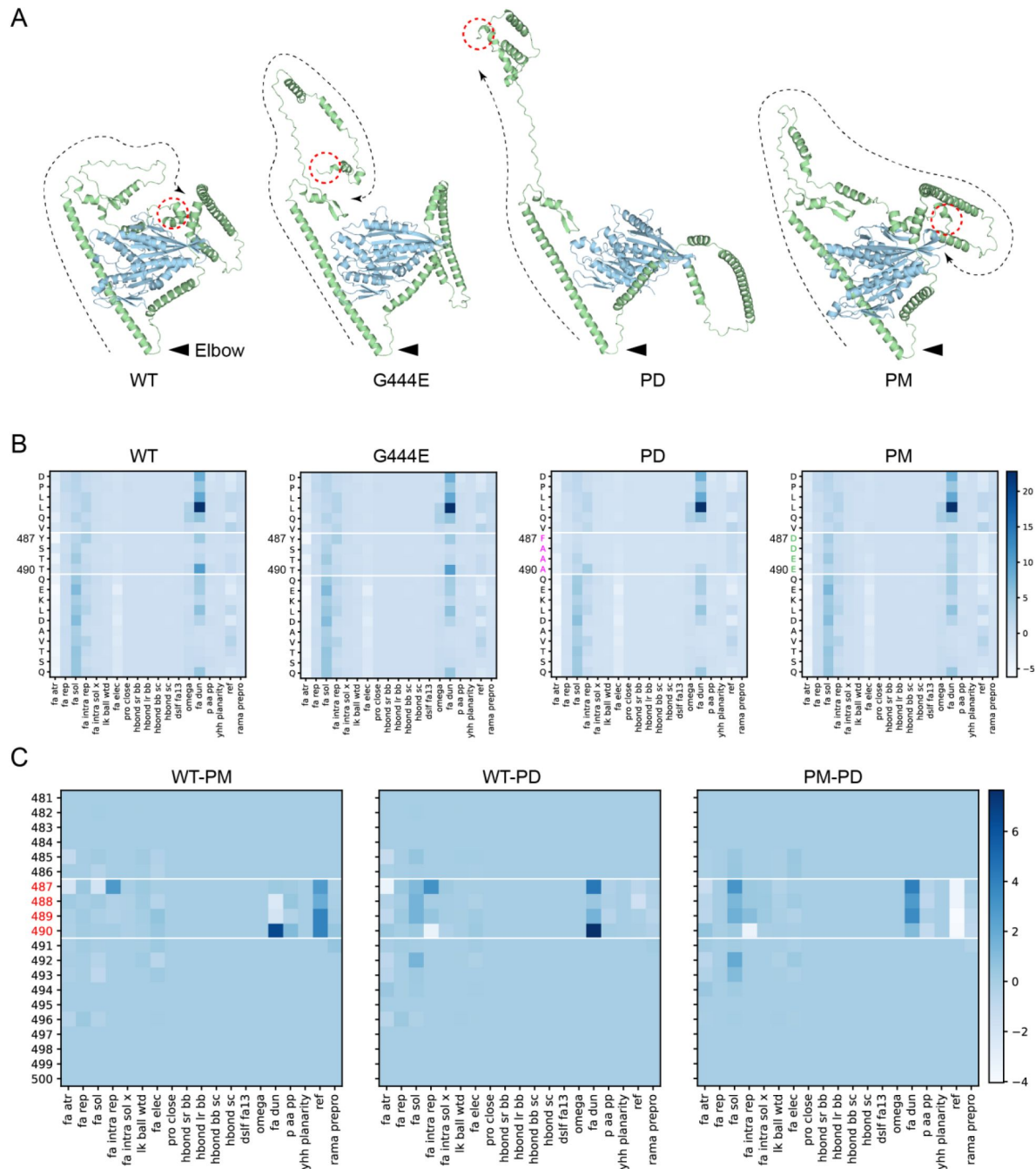


Fig. 3.

Structural models of the OSM-3 kinesin and its mutants

(A) Relaxed structure models of OSM-3 and mutants. Black arrow heads indicate the elbow while red dashed circles mark the C-terminus of the protein. Black dashed lines showed the extending direction from the elbow towards the C-terminus. WT and PM showed close interaction between the tail and motor domain while G444E and PD showed that the tails are far away from the motor. (B) Heatmaps of the energy states of the pre-relaxed structure models from amino acid 481st to 500th, as labeled on the left; the amino acids between the white lines are the elbow region; each row represents an energy item as labeled on the bottom. (C) Heatmaps comparing the energy states by direct subtraction between the mutants and WT. The PD mutant has lower "fa_dun" energy while has higher "ref" energy than that of the PM mutant. Energy terms are explained in Fig. S4.

3 mutants (e.g., G444E) displayed elevated ATPase activity, consistent with structural predictions of autoinhibition release (**Fig. 3A**, **Fig. 4A**) (Xie et al., 2024). While hydrodynamic profiling (e.g., SEC) could further resolve conformational states, our functional assays directly connect predicted structural changes to altered biochemical and cellular activity.

Next, we performed microtubule gliding assays to examine the gliding activity of OSM-3 and its mutants. In support of the changes in ATPase activities, OSM-3PD exhibited increased gliding speed ($0.67 \pm 0.08 \mu\text{m/s}$) compared to WT ($0.52 \pm 0.06 \mu\text{m/s}$) but was slower than OSM-3G444E ($0.87 \pm 0.09 \mu\text{m/s}$) (**Fig. 4B-D**). To examine the processivity of OSM-3, we used a total internal reflection fluorescence (TIRF) microscope to visualize GFP-labeled motors. We did not detect any processive movement of WT OSM-3 or OSM-3PM on microtubules (**Fig. 4B, E-G**). In contrast, OSM-3PD, similar to the hyperactive OSM-3G444E, underwent persistent movement along microtubules, albeit at a slower speed and with a lower landing rate (**Fig. 4B, E-G**). These data indicate that the OSM-3PD mutation disrupts the autoinhibition of full-length OSM-3, providing *in vitro* evidence for the inhibitory effect of NEKL-3 phosphorylation at the OSM-3 elbow region in regulating its autoinhibition.

Genetic suppressor screens identified an intragenic suppressor of OSM-3PD

We performed a forward genetic suppressor screen to identify mutations that could restore the ciliary function of *osm-3PD* (**Fig. 5A**). After screening over 10,000 haploid genomes, we isolated two suppressor mutants that exhibited the uptake of the fluorescent dye DiI through cilia, similar to wild-type animals ($n = 100$ for each) (**Fig. 5B-C**). Both suppressors were intragenic and led to the same A489T missense mutation in the elbow region (**Fig. 5B-C**). Whole-genome sequencing data indicated that these two strains carried distinct mutations in many other loci (Appendix Table S3-4), supporting the notion that these two alleles were isolated independently from our suppressor screens.

We studied the *in vivo* behaviors of the OSM-3PD-A489T variant and observed that the vast majority of GFP fluorescence was localized within cilia (**Fig. 5D**). Notably, neither of the GFP-tagged OSM-3PD-A489T mutants displayed any discernible GFP puncta throughout the entirety of the *C. elegans* body ($N > 100$ for each double mutant), indicating that the A489T mutation completely eliminated the aberrant accumulation of OSM-3PD (**Fig. 5D**). By measuring IFT speeds, we showed that OSM-3PD-A489T moved indistinguishably from WT along the middle and distal ciliary segments (**Fig. 5D-E**, **Fig. S6**). These observations were further confirmed in transgenic animals expressing GFP-tagged OSM-3PD-A489T in the *osm-3(p802)* null allele background (**Fig. 5D-E**, **Fig. S6**).

To probe the regulatory role of T489 phosphorylation, we generated *osm-3(T489E)* and *osm-3(T489A)* mutant animals. Strikingly, both mutants formed axonal puncta (**Fig. S7**), recapitulating the hyperactive phenotype of the OSM-3G444E mutant. While the similar puncta formation in T489E and T489A mutants initially appeared paradoxical, this observation underscores the necessity of dynamic phosphorylation cycling at T489 for proper autoinhibition. Specifically, the T489A mutant likely disrupts phosphorylation-dependent autoinhibition stabilization, leading to constitutive activation, whereas the T489E mutant may mimic a “locked” phosphorylated state, preventing dephosphorylation-dependent release of autoinhibition in cilia and trapping OSM-3 in an aggregation-prone conformation. These results highlight T489 as a structural linchpin whose post-translational modification dynamically regulates motor activity. While the precise molecular mechanism—such as how phosphorylation modulates tail-motor domain interactions—remains to be elucidated, our data conclusively demonstrate that perturbing T489 (even in isolation) destabilizes autoinhibition, driving puncta formation and the constitutive activity.

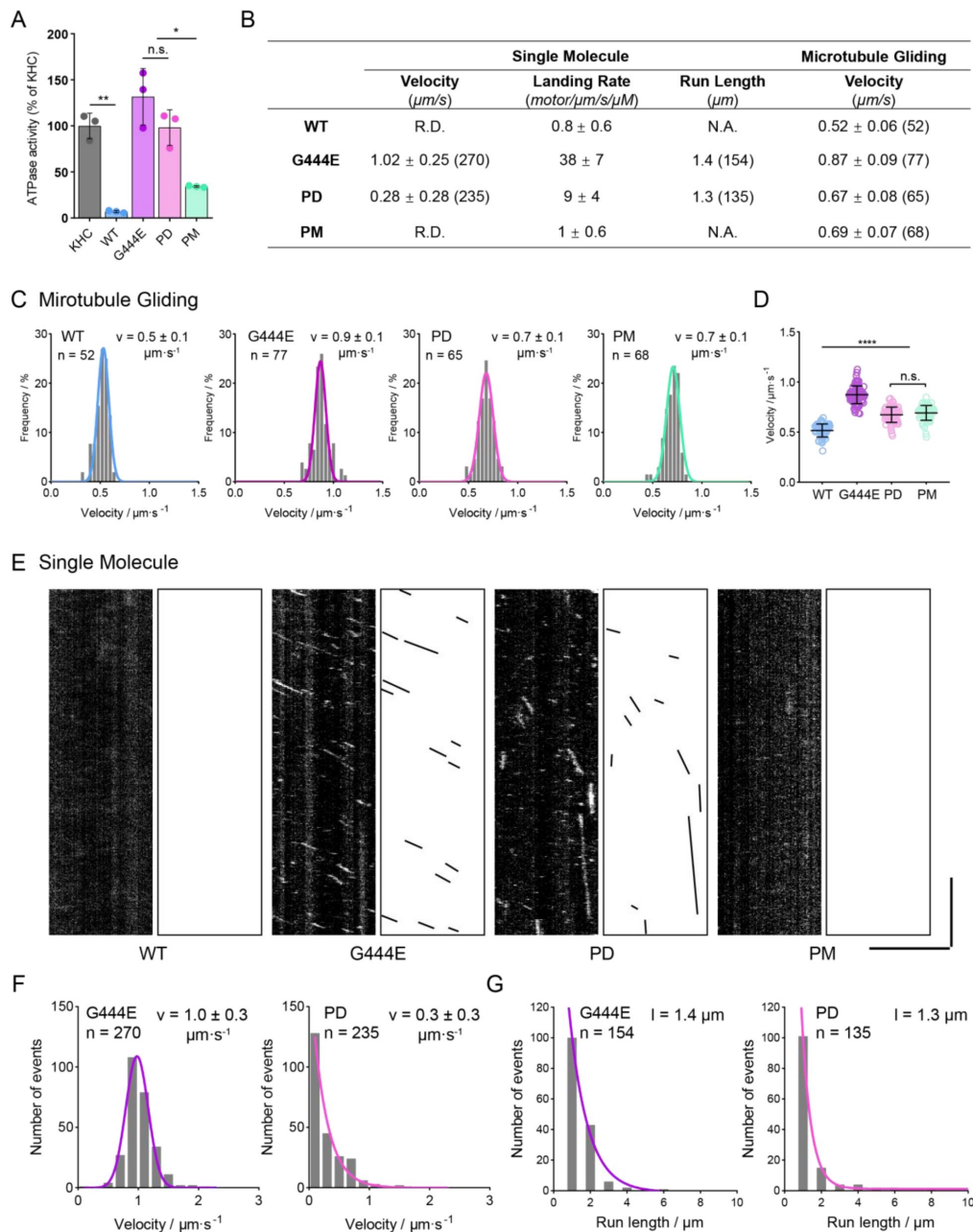


Fig. 4.

Phospho-dead OSM-3 behaves constitutively active in vitro while Phospho-mimic OSM-3 stays autoinhibited.

(A) Microtubule stimulated ATPase activity of WT OSM-3 and mutants. G444E, the hyperactive positive control; KHC, kinesin heavy chain. Average activity of KHC was set to 100% and others was normalized to KHC. (B) Summary of the single-molecular assay and the microtubule gliding assay. R.D., rarely detected. N.A., not available. Data are [mean $\pm$ SD (number of events)]. (C) Velocity distributions of microtubule gliding assays of the indicated OSM-3 constructs. n , total events measured. v , $\mu\text{m/s}^{-1}$, average velocity with standard deviation. (D) Statistics of microtubule gliding velocities shown in (C). (E) Representative kymographs of the single-molecular movements of WT OSM-3 and mutants as indicated. Scale bars, vertical, 10 s; horizontal, 5 μm . (F) Velocity distributions of the single-molecular assays. n , total events measured. v , $\mu\text{m/s}^{-1}$, average velocity with standard deviation. The distribution of G444E was fitted with a Gaussian distribution curve while the distribution of PD was fitted with a one-phase decay curve. (G) Run length distributions of the single-molecular assays. n , total events measured. l , average run length. The curves were fitted with the one-phase decay distribution. *, $p < 0.05$, **, $p < 0.01$, ****, $p < 0.0001$, analyzed by one way ANOVA, p values were adjusted by BH method.

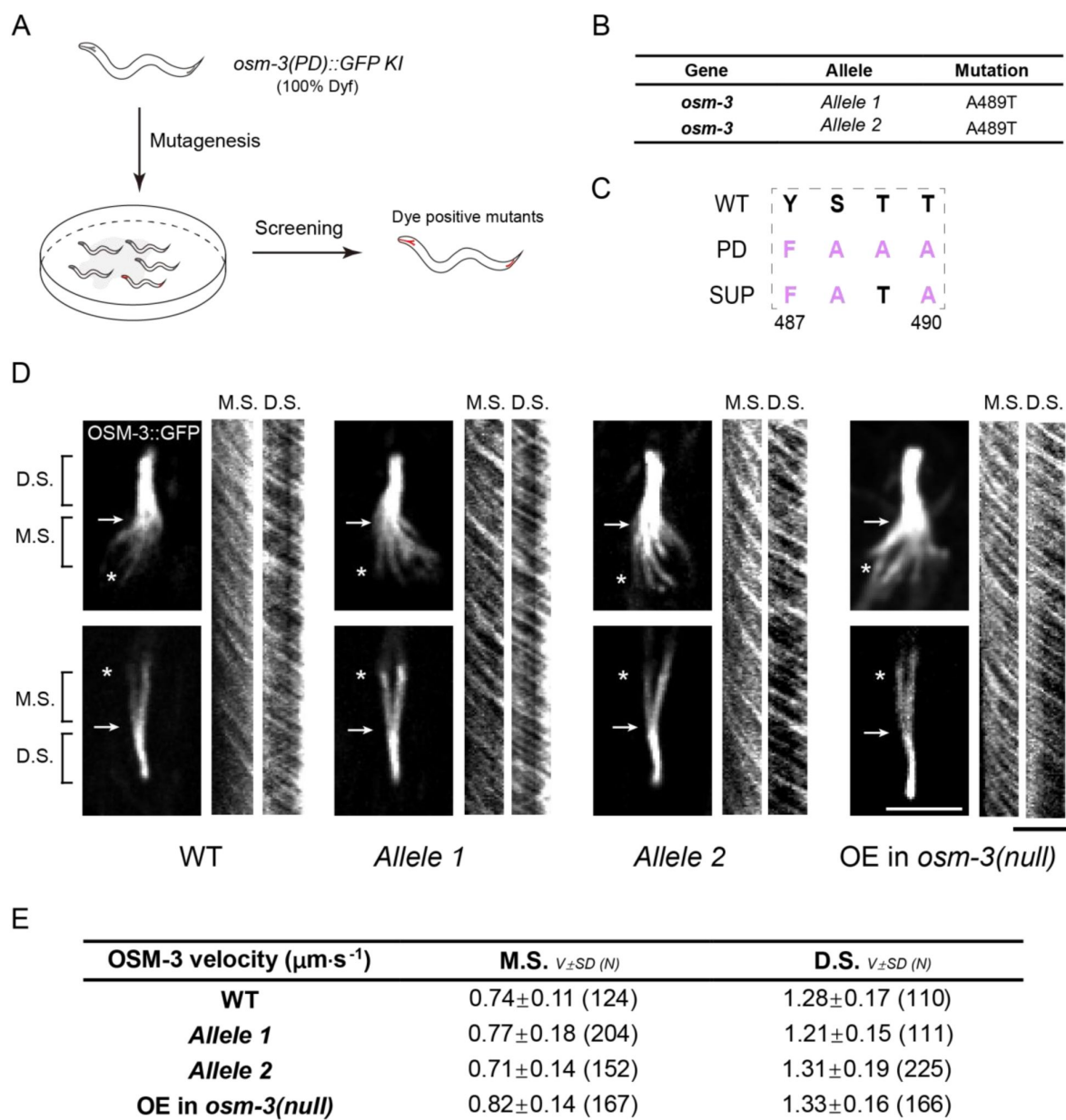


Fig. 5.

Genetic screening identified T489 as the key regulatory residue in the elbow of OSM-3.

(A) Schematics of the forward genetic screen. 100% Dyf *osm-3(PD)::GFP KI* worms were mutated by EMS and F2 progenies were screened for dye filling positive mutants. (B) Two independent suppressor mutants cloned from the genetic screening. (C) Amino acid sequences of the suppressor mutants at the elbow. (D) Representative images of the cilia of the suppressors and the kymographs showing the velocity of OSM-3. The rightmost panel shows the same OSM-3 version (487-489: "FATA") with the suppressors but over-expressed under the ciliary *Pdyf-1* promoter in *osm-3(p802)* worm. The arrows indicate the junction between middle and distal segments while the asterisks indicate the ciliary base. M.S., middle segment; D.S., distal segment. Scale bars, vertical, 10 s; horizontal, 5 μm . (E) Summary of the OSM-3 velocity. M.S., middle segment. D.S. distal segment. Data are [mean $\pm$ SD (number of events)].

Collectively, these *in vivo* findings indicate that the A489T mutation suppresses the defects of OSM-3PD, restoring its normal function and localization.

Discussion

This study introduces a model that elucidates the essential role of OSM-3 elbow phosphorylation in modulating kinesin motility (**Fig. 6**). We demonstrate that NEKL-3 kinase phosphorylates the elbow region of OSM-3 kinesin *in vitro*. Fluorescence of OSM-3::GFP and NEKL-3::GFP was observed in the soma and dendrites of sensory neurons, indicating the sites of protein synthesis and potential regulatory activity. We propose that OSM-3 is synthesized in the soma, where its elbow region is subsequently phosphorylated by NEKL-3. This phosphorylation is critical to inhibit OSM-3 motility prior to its arrival at the cilia. Previous findings have shown that cytoplasmic dynein-1 facilitates the transport of centrioles from the soma along the dendrites to the dendritic tip, the site of ciliary base formation (Li et al., 2017). We hypothesize that inactive OSM-3 is similarly transported, possibly by dynein-1, to the ciliary base as a cargo molecule. Dephosphorylation at the ciliary middle segment appears necessary to activate OSM-3, thereby driving IFT along the ciliary distal segments.

In the absence of phosphorylation at the soma or dendrites, as demonstrated in the OSM-3PD model, OSM-3 may remain constitutively active. This constitutive activity might induce cellular responses that erroneously direct active OSM-3 to neurite tips rather than to cilia. Furthermore, OSM-3 harboring the G444E mutation within the hinge region exhibits constitutive activity (Imanishi et al., 2006; Xie et al., 2024). Our recent study revealed that this hyperactive form of OSM-3 was also absent from cilia and was instead expelled through membrane abscission at the tips of aberrant neurites (Xie et al., 2024). Adjacent glial cells subsequently engulf and degrade the extruded OSM-3G444E, a process mediated by the engulfment receptor CED-1 (Xie et al., 2024). We propose that the constitutively active OSM-3PD is subject to a similar fate, leading to ciliary defects analogous to those observed in the *osm-3* null or *osm-3G444E* alleles.

Our mass spectrometry analysis did not yield conclusive data regarding the specific residue in the elbow region undergoing phosphorylation. However, intriguingly, genetic suppressor screens revealed that the A489T mutation fully restores OSM-3PD localization to the cilia. Although direct mass spectrometric analysis from OSM-3PD-A489T animals is pending, the genetic suppression strongly implicates the role of the T489 site in regulating elbow phosphorylation. Considering the previous proteomic study of *Chlamydomonas* flagellar proteome identified protein phosphatases in cilia (Pazour et al., 2005), we postulate that a protein phosphatase may dephosphorylate inhibitory phosphorylations on the OSM-3 elbow. Once within the cilia, this phosphatase in the ciliary middle segment likely dephosphorylates OSM-3, relieving inhibition from both the motor domain and the elbow region, thereby facilitating OSM-3-driven IFT and contributing to the construction of ciliary distal segments.

In our gliding assays, OSM-3PM has an increased gliding speed of $0.69 \pm 0.07 \mu\text{m/s}$ (**Fig. 4 C-D**), similar to PD mutant. PD and PM mutations are confined to the elbow region, leaving the motor head's mechanochemical properties intact. Upon tail immobilization—which releases autoinhibition—the gliding speeds reflect motor head activity. Single-molecule assays, however, directly resolve their native regulatory states: PD mutants are constitutively active, whereas PM mutants persist in an autoinhibited state (**Fig. 4E-G**). Although monomeric OSM-3 could theoretically mediate single-motor gliding, the previous SEC-MALS data demonstrate that OSM-3 purifies as stable dimers (Xie et al., 2024). Thus, dimeric OSM-3 is perhaps the predominant functional species in our assays.

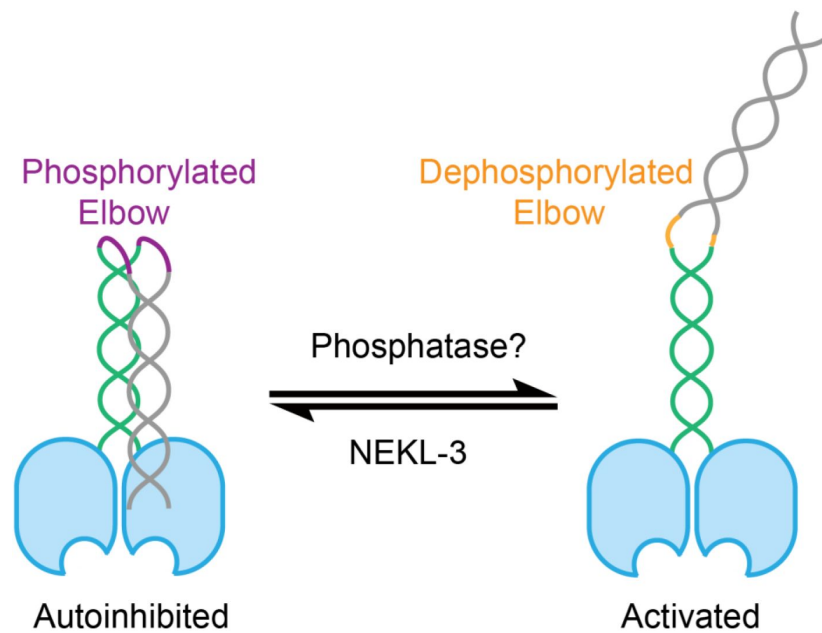


Fig. 6.

Proposed model for elbow regulation of OSM-3 by NEKL-3 phosphorylation

OSM-3 is phosphorylated at the elbow by NEKL-3 and behaved autoinhibited while after dephosphorylation, OSM-3 turns from a compact state to an extended state due to the elbow conformational changes.

As demonstrated in our recent study (Huang et al., JCB, 2025, In press, 10.1083/jcb.202407152), phosphorylation of OSM-3 by NEKL-3 at two distinct regions—Ser96 and the conserved “elbow” motif—differentially regulates its activity and localization. Phosphorylation at Ser96 reduces OSM-3’s ATPase activity and alters its ciliary distribution from the distal segment to a uniform localization, while elbow phosphorylation induces autoinhibition, retaining OSM-3 in the cell body. Strikingly, *in vitro* phosphorylation of OSM-3 by NEKL-3 significantly reduces its microtubule-binding affinity, likely arising from combined modifications at both sites. We propose a model wherein elbow phosphorylation ensures anterograde ciliary transport, while Ser96 phosphorylation fine-tunes distal segment targeting. This multistep regulation may involve distinct phosphatases to reverse phosphorylation at specific sites, a hypothesis warranting further investigation.

It is plausible that additional kinases are involved in elbow phosphorylation. Previous studies have shown NEKL-4 can regulate the stability of ciliary microtubules by affecting tubulin glutamylation, indicating the ciliary functions of the NIMA family kinases (Power et al., 2020 [DOI](#); Power et al., 2024 [DOI](#)). NEKL-3 and its homologue NEKL-4 exhibit similar localization patterns in the soma and dendrites. While NEKL-3 is indispensable for OSM-3 regulation, NEKL-4 appears to have a redundant role in this process. Although our phosphorylation assays utilized recombinant NEKL-3 due to technical constraints that we could not generate recombinant NEKL-4 with activity, it remains possible that NEKL-4, akin to NEKL-3, may also phosphorylate the OSM-3 elbow.

This study primarily elucidates the inhibitory phosphorylation of OSM-3 prior to ciliary entry. Equally intriguing is the regulation of OSM-3 at the ciliary tip, where its activity must be curtailed to halt IFT, thus preventing excessive ciliary elongation. Previous studies have identified DYF-5, DYF-18 kinases and other ciliary kinases as crucial regulators limiting ciliary length across various species (Berman et al., 2003 [DOI](#); Burghoorn et al., 2007 [DOI](#); Maurya & Sengupta, 2021 [DOI](#); Omori et al., 2010 [DOI](#); Ozgul et al., 2011 [DOI](#); Park et al., 2021 [DOI](#); Tucker et al., 2011 [DOI](#); P. Yi et al., 2018 [DOI](#)). It is plausible that elbow phosphorylation by these kinases serves as a regulatory mechanism restraining OSM-3 motility at the ciliary tip. Given the ubiquity of the kinesin elbow region across the superfamily, we propose that phosphorylation regulation of this region may represent a widespread mechanism governing kinesin motility. This suggests a fundamental role for elbow phosphorylation in modulating the activity of kinesins, potentially impacting a broad array of cellular processes.

Methods

C. elegans strain culture

All *C. elegans* strains used in this study were cultured on OP50 seeded NGM (nematode growth medium) plates at 20 °C. Appendix Table S5 summarizes the strains.

Molecular biology

The knock-in *osm-3-pm* and *osm-3-pd* strains were generated by SunyBiotech in the *osm-3::GFP KI* (SYD0199) background by CRSIPR-Cas9 based methods and confirmed by Sanger sequencing.

For OSM-3 transgenic strains, point mutation constructs were generated by PCR and the desired constructs were co-injected with *rol-6[su1006]* into the germ line of young adult hermaphrodites. Marker-positive F1s were singled and the F2s with transgenes were identified as transgenic lines. Two independent lines were maintained and used for experiments.

Live imaging

Live imaging was performed as described (Xie et al., 2024 [↗](#)). Basically, young adult worms were anesthetized with 0.1 mmol/L levamisole in M9 buffer and fixed on a 3% agarose pad. For spinning disk confocal system, imaging was performed on an Axio Observer Z1 microscope (Carl Zeiss, Oberkochen, Germany) equipped with a 100×, 1.49 NA objective, an electron-multiplying (EM) charge-coupled device (CCD) camera (Andor iXon+ DU-897D-C00#BV-500), and the 488/561 nm lines of a Sapphire CW CDRH USB Laser System attached to a spinning disk confocal scan head (Yokogawa CSU-X1 Spinning Disk Unit). Images were acquired by μManager (<https://www.micro-manager.org> [↗](#)) at an exposure time of 200ms and analyzed with ImageJ software.

Structure prediction and energy minimization

OSM-3 model was predicted using LocalColabFold (Evans et al., 2022 [↗](#); Jumper et al., 2021 [↗](#); Mirdita et al., 2022 [↗](#)). Mutated proteins were designed by Pymol 2.6, choosing the rotamer of the mutated residues in G444E, PM and PD models with the least clash as the initial conformation.

To predict mutation-induced conformational changes, the initial models were subjected to Pyrosetta (Chaudhury et al., 2010 [↗](#)). The energies of pre-relaxed models were evaluated with Rosetta Energy Function 2015 (Alford et al., 2017 [↗](#)), and then the relax procedure were applied to the models with default parameters to obtain the relaxed models visualized by Pymol to minimize the energy of these models. In detail, to obtain the relaxed models visualized by Pymol and minimize the energy of these models, the classic relax mover was used in the procedure mentioned above with default settings. The relax script has been uploaded to Github: https://github.com/young55775/RosettaRelax_for_OSM3 [↗](#)

Protein preparation

OSM-3 was expressed in *E. coli* BL21 (DE3) and purified for in vitro assays using established protocols (Imanishi et al., 2006 [↗](#); Xie et al., 2024 [↗](#)). Size-exclusion chromatography coupled with multi-angle light scattering (SEC-MALS) (Xie et al., 2024 [↗](#)) confirmed that recombinant OSM-3 forms a homodimer (173–193 kDa) under physiological conditions, ensuring its dimeric state remained intact.

Basically, point mutations was introduced in to pET.M.3C OSM-3-eGFP-His₆ plasmid for prokaryotic expression. Plasmid transformed *E. coli* (BL21) was cultured at 37°C and induced overnight at 23°C with 0.2 mM IPTG. Cells were lysed in lysis buffer (50 mM NaPO₄ pH8.0, 250 mM NaCl, 20 mM imidazole, 10 mM bME, 0.5 mM ATP, 1 mM MgCl₂, Complete Protease Inhibitor Cocktail (Roche)) and Ni-NTA beads were applied for affinity purification. After incubation, beads were washed with wash buffer (50 mM NaPO₄ pH6.0, 250 mM NaCl, 10 mM bME, 0.1 mM ATP, 1 mM MgCl₂) and eluted with elute buffer (50 mM NaPO₄ pH7.2, 250 mM NaCl, 500 mM imidazole, 10 mM bME, 0.1 mM ATP, 1 mM MgCl₂). Protein concentration was determined by standard Bradford assay.

C elegans *nekl-3* cDNA was cloned in to pGEX-6P GST vector and expressed in *E. coli* BL21 (DE3) and purified for in vitro phosphorylation assays. Plasmid transformed *E. coli* (BL21) was cultured at 37°C and induced overnight at 18°C with 0.5 mM IPTG. Cells were lysed in lysis buffer (50 mM NaPO₄ pH8.0, 250 mM NaCl, 1 mM DTT, Complete Protease Inhibitor Cocktail (Roche)) and GST beads were applied for affinity purification. After incubation, beads were washed with wash buffer (50 mM NaPO₄ pH6.0, 250 mM NaCl, 1 mM DTT) and eluted with elute buffer (50 mM NaPO₄ pH7.2, 150 mM NaCl, 10 mM GSH, 1 mM DTT). Purified proteins were dialyzed against storage buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl). Protein concentration was determined by standard Bradford assay.

ATPase assays

Microtubule stimulated ATPase activity assays were performed with a commercial kit (HTS Kinesin ATPase Endpoint Assay Biochem Kit, Cytoskeleton Inc.) following the manufacturer's instructions.

In vitro Motility assays

Microtubule gliding assays and single molecular assays were performed as described previously (Imanishi et al., 2006 [DOI](#); Xie et al., 2024 [DOI](#)). Briefly, the elution peak fraction was applied to a Zeba™ Spin Desalting Column to exchange the protein into motility buffer (80 mM pipes-K pH6.8, 200 mM KCl, 1 mM EGTA, 2 mM MgCl₂, 0.1 mM ATP, 10 mM bME) before use. For microtubule gliding assays, OSM-3 was flowed into a flow cell in desired concentration and rhodamine labeled microtubules were subsequently flowed-in with assay buffer (BRB80, 1 mM ATP/Mg²⁺, 1% β-Mercaptoethanol, 0.08 mg/mL glucose oxidase, 0.032 mg/mL catalase and 80 mM glucose). For single molecular assays, microtubules were attached to the flow cell surface via antibodies (SAP4G5), and the motors were flowed-in in a desired concentration in assay buffer. The eGFP and the rhodamine were illuminated by 488 or 561 nm laser at 20 mW, and signals were visualized by Olympus IX83 microscopy equipped with a 150×(1.45 NA, oil, Olympus) objective lens and an ORCA-Flash4.0 V3 camera. The system was controlled by Micro-Manager 2.0.

In vitro phosphorylation assay

20 μM purified OSM-3 was incubated with 1 μM GST-NEKL-3 at 30 °C in 100 μL reaction buffer (50 mM Tris-HCl pH 8.0, 10 mM MgCl₂, 150 mM NaCl, and 2 mM ATP) for 30 min. The reaction was terminated by boiling for 5 min with an SDS-sample buffer.

Mass spectrometry

Following NEKL-3 treatment, OSM-3 proteins were resolved by SDS-PAGE and visualized with Coomassie Brilliant Blue staining. Protein bands corresponding to OSM-3 were excised and subjected to digestion using the following protocol: reduction with 5 mM TCEP at 56°C for 30 min; alkylation with 10 mM iodoacetamide in darkness for 45 min at room temperature, and tryptic digestion at 37°C overnight with a 1:20 enzyme-to-protein ratio. The resulting peptides were subjected to mass spectrometry analysis. Briefly, the peptides were analyzed using an UltiMate 3000 RSLCnano system coupled to an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific). We applied an in-house proteome discovery searching algorithm to search the MS/MS data against the *C. elegans* database. Phosphorylation sites were determined using PhosphoRS algorithm with manual validation of MS/MS spectra.

Genetic screening

To isolate suppressors of *osm-3(PD)*, L4 worms were collected in 4 mL M9 and treated with 20 μL ethyl methanesulfonate (EMS) for 4 hours. Adult F2 animals were subjected to Dye-Filling Assay and Dye-positive mutant animals were individually cultured. Dye (DiI, 1,1'-dioctadecyl-3,3,3',3'-tetramethylindocarbocyanine perchlorate; Sigma-Aldrich, St. Louis, MO, USA) was used at a final concentration of 20 μg/mL. Whole-genome sequencing was applied to all suppressor strains to identify candidate genes.

Quantifications and statistical analysis

ImageJ software was used to perform measurements and quantifications of the images. For cilium length, the indicated numbers of phasmid cilia were measured. For IFT velocity, the indicated numbers of IFT particles in amphid and phasmid cilia were randomly selected for the measurement. Microtubule-stimulated ATPase activities were derived from three assays and the average activity of kinesin heavy chain (KHC) was set to 100%. Single molecule and gliding data

were measured using ImageJ software and all the events measured were selected randomly. Statistical analyses were performed in GraphPad Prism. The statistical differences were determined by two-tailed Student's t-test or ANOVA analysis as described in the figure legends. The frequency distribution of IFT velocity was analyzed and fitted with a Gaussian distribution curve.

Data and materials availability

All data are available in the main text or the supplementary materials.

Supplementary figures

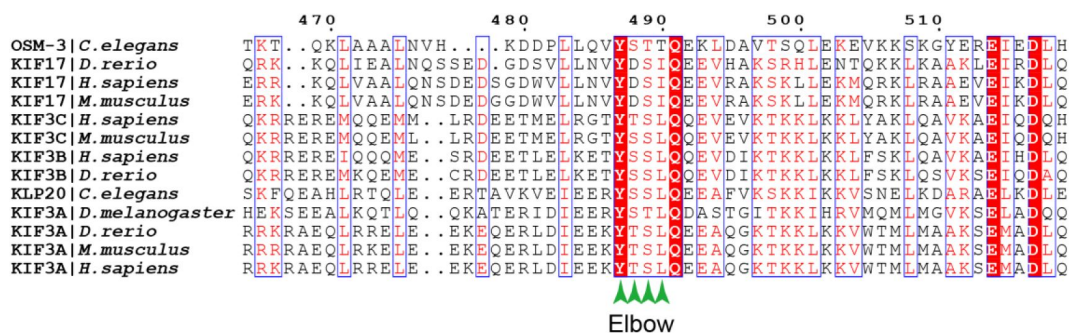
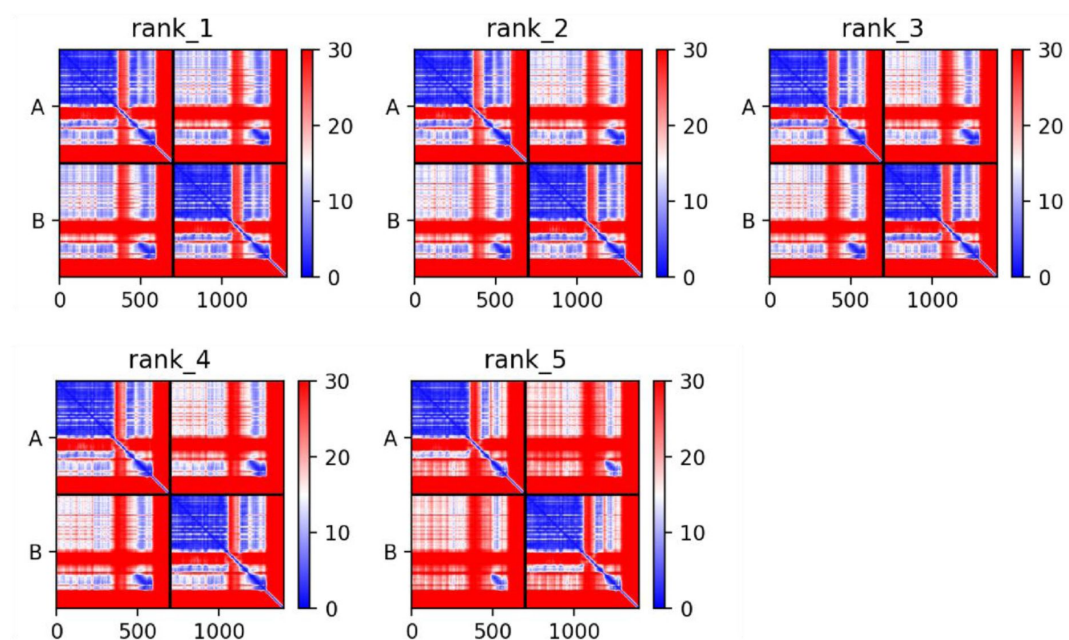


Figure S1.

Kinesin-2 family members have conserved elbow motif

Sequence alignment of OSM-3 and other members in kinesin-2 family (KIF3A, KIF3B, KIF3C and KIF17) from different model organisms as indicated. This figure shows the alignment results around elbow region. Green arrow heads indicate the four phosphorylated residues. The position of amino acids of OSM-3 is labeled at the top of the figure. Sequences were aligned using CLUSTAL O (1.2.4) and presented by ESPript 3.0.

A



B

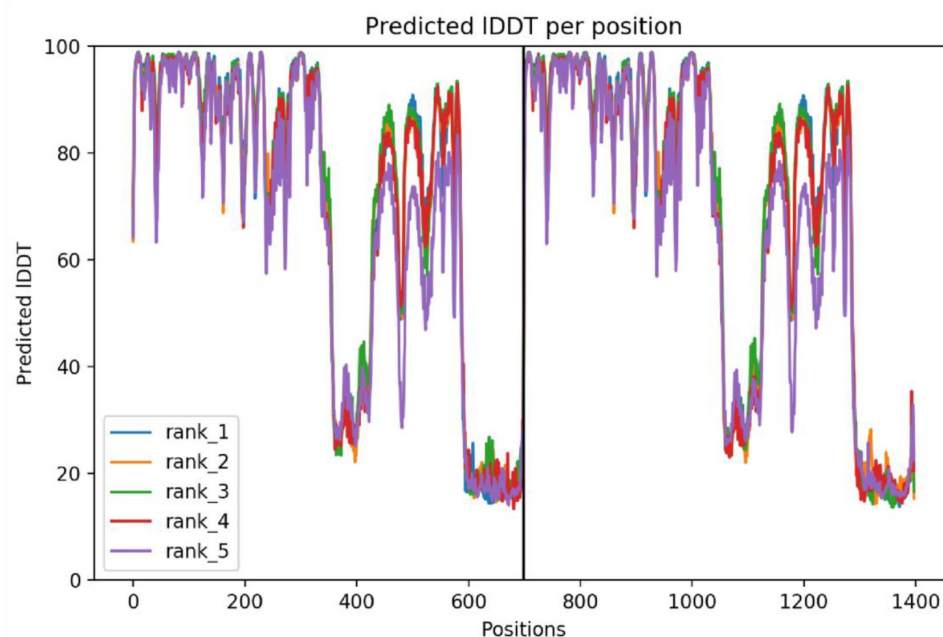


Figure S2.

PAE and pLDDT of OSM-3 dimer models

(A) PAE (Predicted Aligned Error) of OSM-3 dimer presented by LocalColabFold. A total of five predicted model were presented and the “rank_1” model was selected for following analysis. x axis, residue number; y axis, A and B represent for two identical OSM-3 peptide chains. (B) pLDDT (predicted Local Distance Difference Test) for all five predicted models shown in (A).

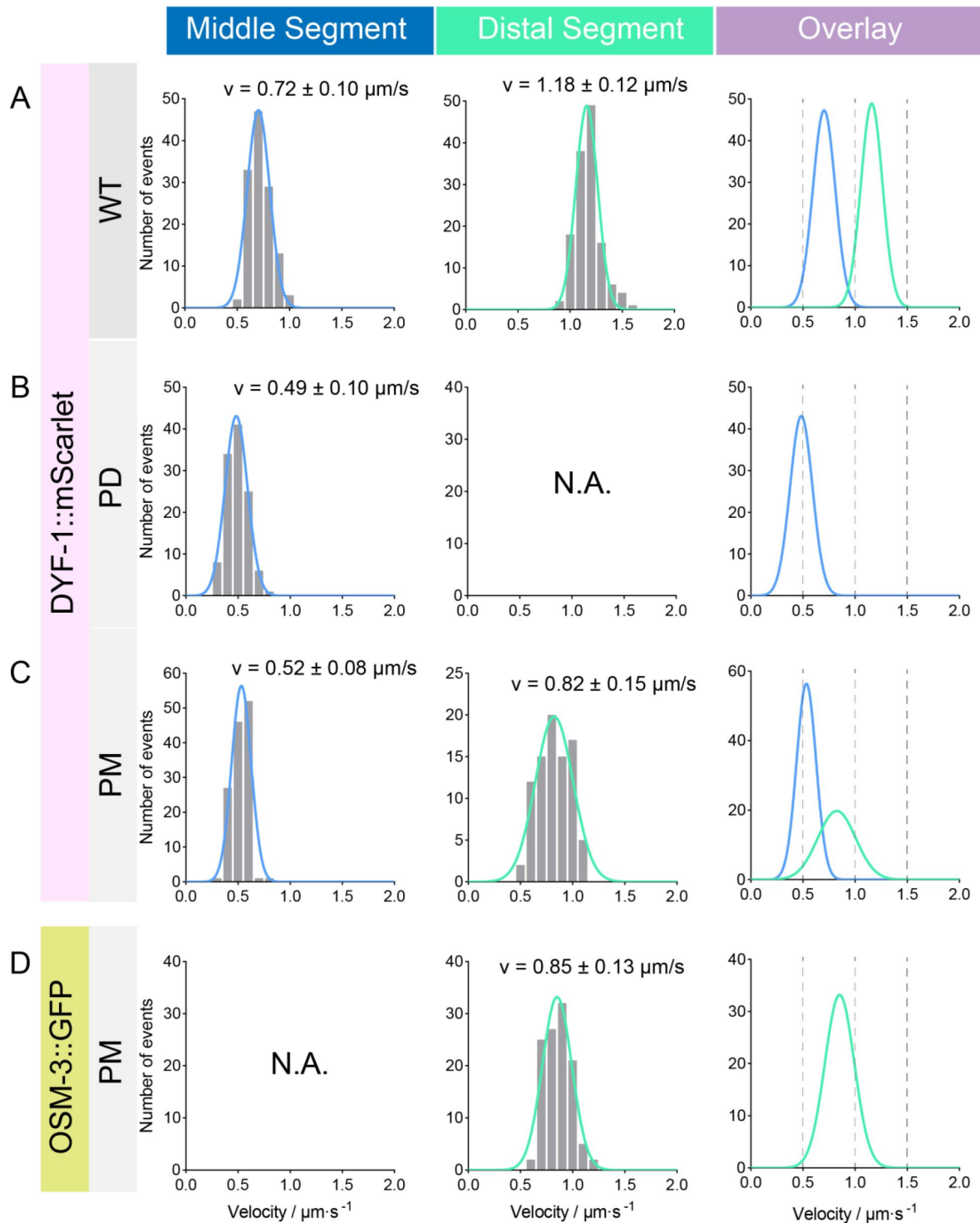


Figure S3.

Statistical analysis of the IFT velocities in *osm-3pd* and *osm-3pm* worms, corresponding to Fig. 2D

(A-C) IFT velocity marked by DYF-1-mScarlet. Genotypes are shown on the left. Left panel, frequency distribution of IFT particles in the middle segment. Middle panel, frequency distribution of IFT particles in the distal segment. Data were fitted with a Gaussian distribution. Right panel, overlay of the fitted curves of middle and distal segments. (D) OSM-3::GFP velocity in *osm-3pm* worms. N.A., not available.

Term	Description
<i>fa_atr</i>	Attractive energy between two atoms on different residues separated by a distance d
<i>fa_rep</i>	Repulsive energy between two atoms on different residues separated by a distance d
<i>fa_sol</i>	Gaussian exclusion implicit solvation energy between protein atoms in different residues
<i>fa_intra_rep</i>	Repulsive energy between two atoms on the same residue separated by a distance d
<i>fa_intra_sol</i>	Gaussian exclusion implicit solvation energy between protein atoms in the same residue
<i>lk_ball_wtd</i>	Orientation-dependent solvation of polar atoms assuming ideal water geometry
<i>fa_elec</i>	Energy of interaction between two nonbonded charged atoms separated by a distance d
<i>pro_close</i>	Penalty for an open proline ring and proline ω bonding energy
<i>hbond_sr_bb</i>	Energy of long-range hydrogen bonds
<i>hbond_lr_bb</i>	Energy of short-range hydrogen bonds
<i>hbond_bb_sc</i>	Energy of backbone–side-chain hydrogen bonds
<i>hbond_sc</i>	Energy of side-chain–side-chain hydrogen bonds
<i>dsf_fa13</i>	Energy of disulfide bridges
<i>omega</i>	Backbone-dependent penalty for cis ω dihedrals that deviate from 0° and trans ω dihedrals that deviate from 180°
<i>fa_dun</i>	Probability that a chosen rotamer is native-like given backbone ϕ , ψ angles
<i>p_aa_pp</i>	Probability of amino acid identity given backbone ϕ , ψ angles
<i>yhh_planarity</i>	Sinusoidal penalty for nonplanar tyrosine χ_3 dihedral angle
<i>ref</i>	Reference energies for amino acid types
<i>rama_prepro</i>	Probability of backbone ϕ , ψ angles given the amino acid type

Figure S4.

Summary of energy terms in Rosetta Energy Function, corresponding to Fig.3

(B-C) This table explains the terms used in the analysis in Fig.3 (B-C). Modified from Alford *et al.*, *J Chem. Theory. Comput.*, 2017 (Alford *et al.*, 2017).

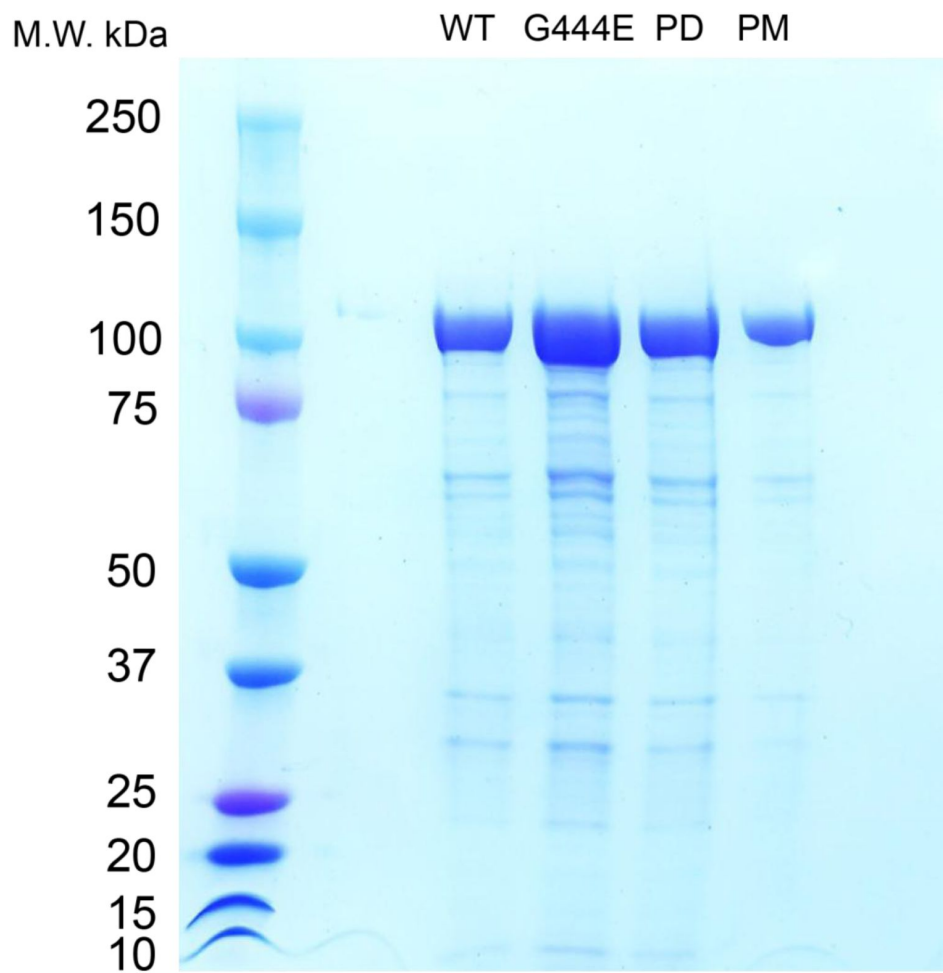


Figure S5.

SDS-PAGE of purified recombinant OSM-3 mutants

SDS-PAGE of the purified recombinant OSM-3::eGFP and mutants, Coomassie Blue stained. The elution peaks of each recombinant protein were shown, labeled on the top.

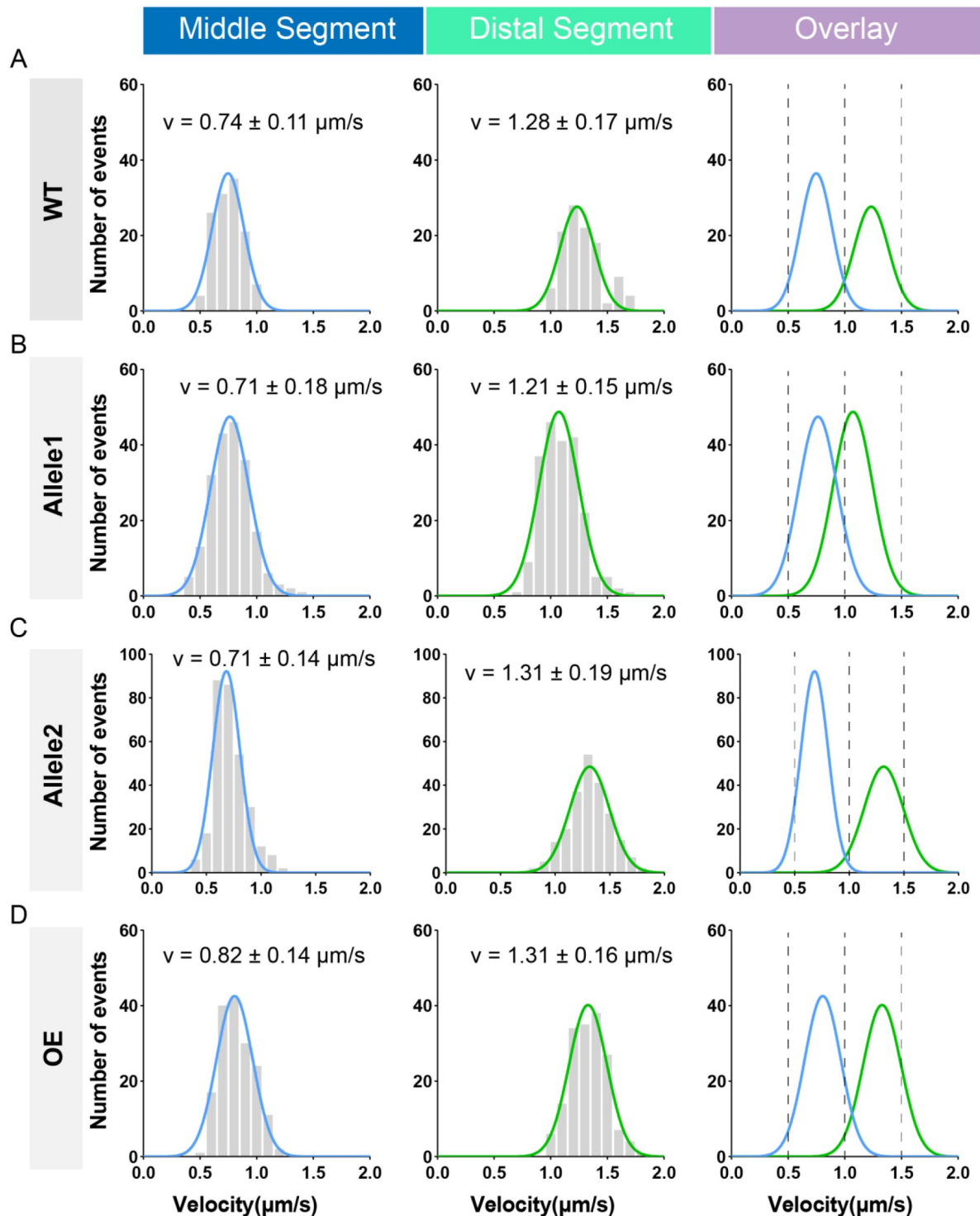


Figure S6.

Statistical analysis of the IFT velocities in *osm-3pd* suppressors, corresponding to Fig. 5E

(A-C) OSM-3 velocity of WT and mutant proteins marked by GFP. Genotypes are shown on the left. Left panel, frequency distribution of IFT particles in the middle segment. Middle panel, frequency distribution of IFT particles in the distal segment. Data were fitted with the Gaussian distribution. Right panel, overlay of the fitted curves of middle and distal segments. (D) Velocity and frequency distribution of overexpressed OSM-3 (487-489: "FATA") under the ciliary *Pdyf-1* promoter in *osm-3(p802)* worm.

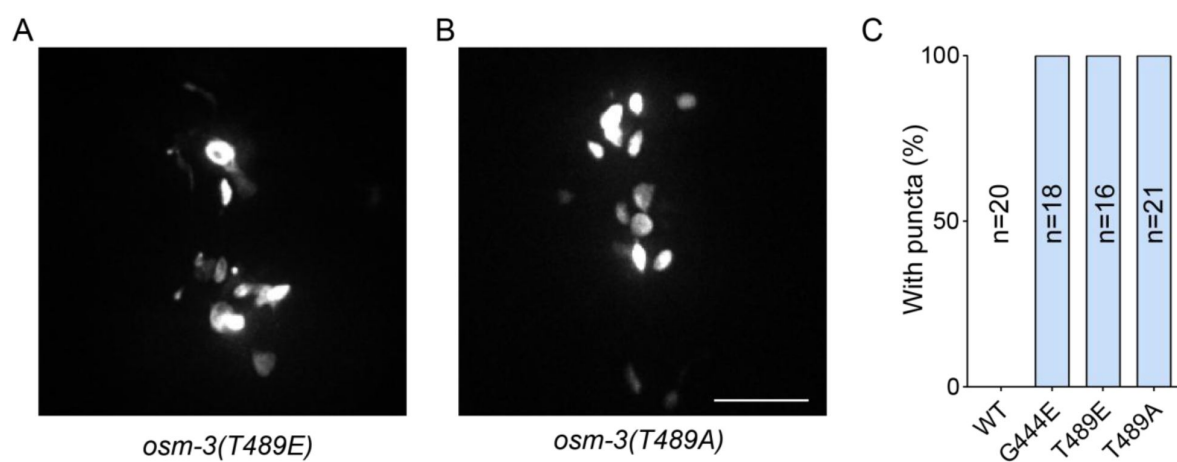


Figure S7.

T489E and T489A of OSM-3 cause aggregated signals.

(A-B) Representative images of OSM-3(T489E) and OSM-3(T489A) overexpressed in *osm-3(p802)* strain under the ciliary *Pdyf-1* promoter, respectively. The images showed the areas around nerve rings. Scale bar, 10 μm . (C) Statistics of strains in (A-B), showing the percentage of worms with bright puncta around nerve rings for each strain. n, number of worms checked.

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Additional information

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Author contributions

G. Ou, G. Chen, Z. Zhu, S. Xie and T. Zhou conceived the project. G. Chen performed the experiments, collected and analyzed the data under the supervision of G. Ou. Z. Zhu prepared recombinant NEKL-3 and performed OSM-3 phosphorylation experiments. Z. Guo performed the energy minimization experiments. G. Ou and G. Chen wrote the manuscript and others revised the manuscript.

Additional files

Appendix Table S1 [↗](#)

Appendix Table S2 [↗](#)

Appendix Table S3 [↗](#)

Appendix Table S4 [↗](#)

Appendix Table S5 [↗](#)

References

- Alford R. F., Leaver-Fay A., Jeliazkov J. R., O'Meara M. J., DiMaio F. P., Park H., Shapovalov M. V., Renfrew P. D., Mulligan V. K., Kappel K., Labonte J. W., Pacella M. S., Bonneau R., Bradley P., Dunbrack R. L., Das R., Baker D., Kuhlman B., Kortemme T., Gray J. J (2017) **The Rosetta All-Atom Energy Function for Macromolecular Modeling and Design** *J Chem Theory Comput* **13**:3031–3048 <https://doi.org/10.1021/acs.jctc.7b00125>
- Anvarian Z., Mykytyn K., Mukhopadhyay S., Pedersen L. B., Christensen S. T (2019) **Cellular signalling by primary cilia in development, organ function and disease** *Nat Rev Nephrol* **15**:199–219 <https://doi.org/10.1038/s41581-019-0116-9>
- Banerjee R., Chakraborty P., Yu M. C., Gunawardena S (2021) **A stop or go switch: glycogen synthase kinase 3 β phosphorylation of the kinesin 1 motor domain at Ser314 halts motility without detaching from microtubules** *Development* **148** <https://doi.org/10.1242/dev.199866>
- Barstead R., Moulder G., Cobb B., Frazee S., Henthorn D., Holmes J., Jerebie D., Landsdale M., Osborn J., Pritchett C., Robertson J., Rummage J., Stokes E., Vishwanathan M., Mitani S., Gengyo-Ando K., Funatsu O., Hori S., Imae R., ..., Consorti C. E. D. M (2012) **Large-Scale Screening for Targeted Knockouts in the Genome** *G3-Genes Genomes Genetics* **2**:1415–1425 <https://doi.org/10.1534/g3.112.003830>
- Berman S. A., Wilson N. F., Haas N. A., Lefebvre P. A (2003) **A novel MAP kinase regulates flagellar length in Chlamydomonas** *Curr Biol* **13**:1145–1149 [https://doi.org/10.1016/s0960-9822\(03\)00415-9](https://doi.org/10.1016/s0960-9822(03)00415-9)
- Brady S. T., Morfini G. A (2017) **Regulation of motor proteins, axonal transport deficits and adult-onset neurodegenerative diseases** *Neurobiology of Disease* **105**:273–282 <https://doi.org/10.1016/j.nbd.2017.04.010>
- Burghoorn J., Dekkers M. P., Rademakers S., de Jong T., Willemsen R., Jansen G. (2007) **Mutation of the MAP kinase DYF-5 affects docking and undocking of kinesin-2 motors and reduces their speed in the cilia of Caenorhabditis elegans** *Proc Natl Acad Sci U S A* **104**:7157–7162 <https://doi.org/10.1073/pnas.0606974104>
- Burute M., Kapitein L. C (2019) **Cellular Logistics: Unraveling the Interplay Between Microtubule Organization and Intracellular Transport** *Annu Rev Cell Dev Biol* **35**:29–54 <https://doi.org/10.1146/annurev-cellbio-100818-125149>
- Cason S. E., Holzbaur E. L. F (2022) **Selective motor activation in organelle transport along axons** *Nat Rev Mol Cell Biol* **23**:699–714 <https://doi.org/10.1038/s41580-022-00491-w>
- Chaudhury S., Lyskov S., Gray J. J (2010) **PyRosetta: a script-based interface for implementing molecular modeling algorithms using Rosetta** *Bioinformatics* **26**:689–691 <https://doi.org/10.1093/bioinformatics/btq007>
- Chivukula R. R., Montoro D. T., Leung H. M., Yang J., Shamseldin H. E., Taylor M. S., Dougherty G. W., Zariwala M. A., Carson J., Daniels M. L. A., Sears P. R., Black K. E., Hariri L. P., Almogharri I., Frenkel E. M., Vinarsky V., Omran H., Knowles M. R., Tearney G. J., ..., Sabatini D. M. (2020) **A**

human ciliopathy reveals essential functions for NEK10 in airway mucociliary clearance *Nat Med* **26**:244–251 <https://doi.org/10.1038/s41591-019-0730-x>

Christensen J. R., Reck-Peterson S. L. (2022) **Hitchhiking Across Kingdoms: Cotransport of Cargos in Fungal, Animal, and Plant Cells** *Annu Rev Cell Dev Biol* **38**:155–178 <https://doi.org/10.1146/annurev-cellbio-120420-104341>

Coy D. L., Hancock W. O., Wagenbach M., Howard J. (1999) **Kinesin's tail domain is an inhibitory regulator of the motor domain** *Nat Cell Biol* **1**:288–292 <https://doi.org/10.1038/13001>

DeBerg H. A., Blehm B. H., Sheung J., Thompson A. R., Bookwalter C. S., Torabi S. F., Schroer T. A., Berger C. L., Lu Y., Trybus K. M., Selvin P. R. (2013) **Motor Domain Phosphorylation Modulates Kinesin-1 Transport** *Journal of Biological Chemistry* **288**:32612–32621 <https://doi.org/10.1074/jbc.M113.515510>

Dietrich K. A., Sindelar C. V., Brewer P. D., Downing K. H., Cremo C. R., Rice S. E. (2008) **The kinesin-1 motor protein is regulated by a direct interaction of its head and tail** *Proc Natl Acad Sci U S A* **105**:8938–8943 <https://doi.org/10.1073/pnas.0803575105>

Espeut J., Gaussen A., Bieling P., Morin V., Prieto S., Fesquet D., Surrey T., Abrieu A. (2008) **Phosphorylation relieves autoinhibition of the kinetochore motor Cenp-E** *Mol Cell* **29**:637–643 <https://doi.org/10.1016/j.molcel.2008.01.004>

Evans R., O'Neill M., Pritzel A., Antropova N., Senior A., Green T., Židek A., Bates R., Blackwell S., Yim J., Ronneberger O., Bodenstein S., Zielinski M., Bridgland A., Potapenko A., Cowie A., Tunyasuvunakool K., Jain R., Clancy E., ..., Hassabis D. (2022) **Protein complex prediction with AlphaFold-Multimer** *bioRxiv* :2021.2010.2004.463034 <https://doi.org/10.1101/2021.10.04.463034>

Friedman D. S., Vale R. D. (1999) **Single-molecule analysis of kinesin motility reveals regulation by the cargo-binding tail domain** *Nat Cell Biol* **1**:293–297 <https://doi.org/10.1038/13008>

Fry A. M., O'Regan L., Sabir S. R., Bayliss R. (2012) **Cell cycle regulation by the NEK family of protein kinases** *J Cell Sci* **125**:4423–4433 <https://doi.org/10.1242/jcs.111195>

Hirokawa N., Noda Y., Tanaka Y., Niwa S. (2009) **Kinesin superfamily motor proteins and intracellular transport** *Nat Rev Mol Cell Biol* **10**:682–696 <https://doi.org/10.1038/nrm2774>

Imanishi M., Endres N. F., Gennerich A., Vale R. D. (2006) **Autoinhibition regulates the motility of the C. elegans intraflagellar transport motor OSM-3** *J Cell Biol* **174**:931–937 <https://doi.org/10.1083/jcb.200605179>

Ishikawa H., Marshall W. F. (2011) **Ciliogenesis: building the cell's antenna** *Nat Rev Mol Cell Biol* **12**:222–234 <https://doi.org/10.1038/nrm3085>

Jumper J., Evans R., Pritzel A., Green T., Figurnov M., Ronneberger O., Tunyasuvunakool K., Bates R., Židek A., Potapenko A., Bridgland A., Meyer C., Kohl S. A. A., Ballard A. J., Cowie A., Romera-Paredes B., Nikolov S., Jain R., Adler J., ..., Hassabis D. (2021) **Highly accurate protein structure prediction with AlphaFold** *Nature* **596**:583–589 <https://doi.org/10.1038/s41586-021-03819-2>

- Klena N., Pigino G (2022) **Structural Biology of Cilia and Intraflagellar Transport** *Annu Rev Cell Dev Biol* <https://doi.org/10.1146/annurev-cellbio-120219-034238>
- Li W., Yi P., Zhu Z., Zhang X., Li W., Ou G (2017) **Centriole translocation and degeneration during ciliogenesis in *Caenorhabditis elegans* neurons** *The EMBO Journal* **36**:2553–2566 <https://doi.org/10.15252/emboj.201796883>
- Liang Y., Pang Y., Wu Q., Hu Z., Han X., Xu Y., Deng H., Pan J (2014) **FLA8/KIF3B Phosphorylation Regulates Kinesin-II Interaction with IFT-B to Control IFT Entry and Turnaround** *Developmental Cell* **30**:585–597 <https://doi.org/10.1016/j.devcel.2014.07.019>
- Maurya A. K., Sengupta P (2021) **xbx-4, a homolog of the Joubert syndrome gene FAM149B1, acts via the CCRK and RCK kinase cascade to regulate cilia morphology** *Current Biology* **31**:5642–5649 <https://doi.org/10.1016/j.cub.2021.10.027>
- Mirdita M., Schütze K., Moriwaki Y., Heo L., Ovchinnikov S., Steinegger M (2022) **ColabFold: making protein folding accessible to all** *Nat Methods* **19**:679–682 <https://doi.org/10.1038/s41592-022-01488-1>
- Mitra A., Loseva E., Peterman E. J. G (2024) **IFT cargo and motors associate sequentially with IFT trains to enter cilia of *C. elegans*** *Nature Communications* **15** <https://doi.org/10.1038/s41467-024-47807-2>
- Nachury M. V., Mick D. U (2019) **Establishing and regulating the composition of cilia for signal transduction** *Nat Rev Mol Cell Biol* **20**:389–405 <https://doi.org/10.1038/s41580-019-0116-4>
- Omori Y., Chaya T., Katoh K., Kajimura N., Sato S., Muraoka K., Ueno S., Koyasu T., Kondo M., Furukawa T (2010) **Negative regulation of ciliary length by ciliary male germ cell-associated kinase (Mak) is required for retinal photoreceptor survival** *Proc Natl Acad Sci U S A* **107**:22671–22676 <https://doi.org/10.1073/pnas.1009437108>
- Ou G., Blacque O., Snow J. J., Leroux M. R., Scholey J. M. (2005) **Functional coordination of intraflagellar transport motors** *Nature* **436**:583–587 <https://doi.org/10.1038/nature03818>
- Ou G., Scholey J. M (2022) **Motor Cooperation During Mitosis and Ciliogenesis** *Annu Rev Cell Dev Biol* <https://doi.org/10.1146/annurev-cellbio-121420-100107>
- Ou G., Scholey J. M (2022) **Motor Cooperation During Mitosis and Ciliogenesis** *Annual Review of Cell and Developmental Biology* **38**:49–74 <https://doi.org/10.1146/annurev-cellbio-121420-100107>
- Ozgul R. K., Siemiatkowska A. M., Yucel D., Myers C. A., Collin R. W., Zonneveld M. N., Beryozkin A., Banin E., Hoyng C. B., van den Born L. I., European Retinal Disease C., Bose R., Shen W., Sharon D., Cremers F. P., Klevering B. J., den Hollander A. I., Corbo J. C. (2011) **Exome sequencing and cis-regulatory mapping identify mutations in MAK, a gene encoding a regulator of ciliary length, as a cause of retinitis pigmentosa** *Am J Hum Genet* **89**:253–264 <https://doi.org/10.1016/j.ajhg.2011.07.005>
- Park K., Li C., Tsiropoulou S., Gonçalves J., Kondratev C., Pelletier L., Blacque O. E., Leroux M. R (2021) **CDKL kinase regulates the length of the ciliary proximal segment** *Curr Biol* **31**:2359–2373 <https://doi.org/10.1016/j.cub.2021.03.068>

- Pazour G. J., Agrin N., Leszyk J., Witman G. B (2005) **Proteomic analysis of a eukaryotic cilium** *J Cell Biol* **170**:103–113 <https://doi.org/10.1083/jcb.200504008>
- Power K. M., Akella J. S., Gu A., Walsh J. D., Bellotti S., Morash M., Zhang W., Ramadan Y. H., Ross N., Golden A., Smith H. E., Barr M. M., O'Hagan R (2020) **Mutation of NEKL-4/NEK10 and TTL genes suppress neuronal ciliary degeneration caused by loss of CCP-1 deglutamylase function** *PLOS Genetics* **16** <https://doi.org/10.1371/journal.pgen.1009052>
- Power K. M., Nguyen K. C., Silva A., Singh S., Hall D. H., Rongo C., Barr M. M (2024) **NEKL-4 regulates microtubule stability and mitochondrial health in ciliated neurons** *Journal of Cell Biology* **223** <https://doi.org/10.1083/jcb.202402006>
- Prevo B., Mangeol P., Oswald F., Scholey J. M., Peterman E. J. G (2015) **Functional differentiation of cooperating kinesin-2 motors orchestrates cargo import and transport in. cilia** *Nature Cell Biology* **17**:1536–1545 <https://doi.org/10.1038/ncb3263>
- Prevo B., Scholey J. M., Peterman E. J. G (2017) **Intraflagellar transport: mechanisms of motor action, cooperation, and cargo delivery** *Febs J* **284**:2905–2931 <https://doi.org/10.1111/febs.14068>
- Sato-Yoshitake R., Yorifuji H., Inagaki M., Hirokawa N (1992) **The phosphorylation of kinesin regulates its binding to synaptic vesicles** *Journal of Biological Chemistry* **267**:23930–23936 [https://doi.org/10.1016/s0021-9258\(18\)35926-x](https://doi.org/10.1016/s0021-9258(18)35926-x)
- Schäfer B., Götz C., Montenarh M (2008) **The kinesin I family member KIF5C is a novel substrate for protein kinase CK2** *Biochemical and Biophysical Research Communications* **375**:179–183 <https://doi.org/10.1016/j.bbrc.2008.07.107>
- Sleigh J. N., Rossor A. M., Fellows A. D., Tosolini A. P., Schiavo G (2019) **Axonal transport and neurological disease** *Nat Rev Neurol* **15**:691–703 <https://doi.org/10.1038/s41582-019-0257-2>
- Smith L. A., Bukanov N. O., Husson H., Russo R. J., Barry T. C., Taylor A. L., Beier D. R., Ibraghimov-Beskrovnaya O (2006) **Development of polycystic kidney disease in juvenile cystic kidney mice: insights into pathogenesis, ciliary abnormalities, and common features with human disease** *J Am Soc Nephrol* **17**:2821–2831 <https://doi.org/10.1681/asn.2006020136>
- Tan Z., Yue Y., Leprevost F., Haynes S., Basrur V., Nesvizhskii A. I., Verhey K. J., Cianfrocco M. A (2023) **Autoinhibited kinesin-1 adopts a hierarchical folding pattern** *eLife* **12** <https://doi.org/10.7554/eLife.86776>
- Taschner M., Lorentzen E (2016) **The Intraflagellar Transport Machinery** *Cold Spring Harb Perspect Biol* **8** <https://doi.org/10.1101/cshperspect.a028092>
- Thiel C., Kessler K., Giessel A., Dimmler A., Shalev S. A., von der Haar S., Zenker M., Zahnleiter D., Stöss H., Beinder E., Abou Jamra R., Ekici A. B., Schröder-Kress N., Aigner T., Kirchner T., Reis A., Brandstätter J. H., Rauch A. (2011) **NEK1 mutations cause short-rib polydactyly syndrome type majewski** *Am J Hum Genet* **88**:106–114 <https://doi.org/10.1016/j.ajhg.2010.12.004>
- Tucker B. A., Scheetz T. E., Mullins R. F., DeLuca A. P., Hoffmann J. M., Johnston R. M., Jacobson S. G., Sheffield V. C., Stone E. M (2011) **Exome sequencing and analysis of induced pluripotent stem cells identify the cilia-related gene male germ cell-associated kinase (MAK) as a cause of retinitis pigmentosa** *Proc Natl Acad Sci U S A* **108**:E569–576 <https://doi.org/10.1073/pnas.1108918108>

Verhey K. J., Hammond J. W (2009) **Traffic control: regulation of kinesin motors** *Nat Rev Mol Cell Biol* **10**:765–777 <https://doi.org/10.1038/nrm2782>

Weijman J. F., Yadav S. K. N., Surridge K. J., Cross J. A., Borucu U., Mantell J., Woolfson D. N., Schaffitzel C., Dodding M. P (2022) **Molecular architecture of the autoinhibited kinesin-1 lambda particle** *Sci Adv* **8**:eabp9660 <https://doi.org/10.1126/sciadv.abp9660>

Xie C., Chen G., Li M., Huang P., Chen Z., Lei K., Li D., Wang Y., Cleetus A., Mohamed M. A., Sonar P., Feng W., Ökten Z., Ou G (2024) **Neurons dispose of hyperactive kinesin into glial cells for clearance** *Embo j* **43**:2606–2635 <https://doi.org/10.1038/s44318-024-00118-0>

Yi P., Xie C., Ou G (2018) **The kinases male germ cell-associated kinase and cell cycle-related kinase regulate kinesin-2 motility in Caenorhabditis elegans neuronal cilia** *Traffic* **19**:522–535 <https://doi.org/10.1111/tra.12572>

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Reviewer #1 (Public review):

Summary:

This manuscript is a focused investigation of the phosphor-regulation of a *C. elegans* kinesin-2 motor protein, OSM-3. In *C. elegans* sensory ciliary, kinesin-2 motor proteins Kinesin-II complex and OSM-3 homodimer transport IFT trains anterogradely to the ciliary tip. Kinesin-II carries OSM-3 as an inactive passenger from the ciliary base to the middle segment, where kinesin-II dissociates from IFT trains and OSM-3 gets activated and transports IFT trains to the distal segment. Therefore, activation/inactivation of OSM-3 plays an essential role in its ciliary function.

Strengths:

In this study, using mass spectrometry, the authors have shown that the NEKL-3 kinase phosphorylates a serine/threonine patch at the hinge region between coiled coils 1 and 2 of an OSM-3 dimer, referred to as the elbow region in ubiquitous kinesin-1. Phosphomimic mutants of these sites inhibit OSM-3 motility both in vitro and in vivo, suggesting that this phosphorylation is critical for the autoinhibition of the motor. Conversely, phospho-dead mutants of these sites hyperactivate OSM-3 motility in vitro and affect the localization of OSM3 in *C. elegans*. The authors also showed that Alanine to Tyrosine mutation of one of the phosphorylation rescues OS-3 function in live worms.

Weaknesses:

Collectively, this study presents evidence for the physiological role of OSM-3 elbow phosphorylation in its autoregulation, which affects ciliary localization and function of this motor. Overall, the work is well performed, and the results mostly support the conclusions of this manuscript. During revision, the authors further supported conclusions and ruled out alternative explanations by filling some logical gaps with new experimental evidence and in-text clarifications.

Comments on revisions: I have no additional comments or concerns.

<https://doi.org/10.7554/eLife.103648.2.sa2>

Reviewer #2 (Public review):

Summary:

The regulation of kinesin is fundamental to cellular morphogenesis. Previously, it has been shown that OSM-3, a kinesin required for intraflagellar transport (IFT), is regulated by

autoinhibition. However, it remains totally elusive how the autoinhibition of OSM-3 is released. In this study, the authors have shown that NEKL-3 phosphorylates OSM-3 and release its autoinhibition.

The authors found NEKL-3 directly phosphorylates OSM-3 (Figure 1). The phosphorylated residue is the "elbow" of OSM-3. The authors introduced phospho-dead (PD) and phospho-mimic (PM) mutations by genome editing and found that the OSM-3(PD) protein does not form cilia, and instead, accumulates to the axonal tips. The phenotype is similar to another constitutive active mutant of OSM-3, OSM-3(G444A) (Imanishi et al., 2006; Xie et al., 2024). *osm-3*(PM) has shorter cilia, which resembles with loss of function mutants of *osm-3* (Figure 2). The authors did structural prediction and shows that G444E and PD mutations change the conformation of OSM-3 protein (Figure 3). In the single molecule assays G444E and PD mutations exhibited increased landing rate (Figure 4). By unbiased genetic screening, the authors identified a suppressor mutant of *osm-3*(PD), in which A489T occurs. The result confirms the importance of this residue. Based on these results, the authors suggest that NEKL-3 induces phosphorylation of the elbow domain and inactivates OSM-3 motor when the motor is synthesized in the cell body. This regulation is essential for the proper cilia formation.

Strengths:

The finding is interesting and gives new insight into how IFT motor is regulated.

Comments on revisions: In the revised manuscript, the authors describe why they focused on NEKL-3 and detailed experimental procedures are presented.

My only minor concern is the title, which appears to be too general. Researchers in the motor protein field may firstly assume this paper focuses on kinesin-1, because the "elbow" domain was originally suggested in kinesin-1. This paper newly determines the elbow region of OSM-3 and shows its crucial role in autoinhibition. Therefore, a more specific title, "Kinesin-2 Autoinhibition Requires Elbow Phosphorylation" or "OSM-3 Autoinhibition Requires Elbow phosphorylation" may be better.

<https://doi.org/10.7554/eLife.103648.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript is a focused investigation of the phosphor-regulation of a C. elegans kinesin-2 motor protein, OSM-3. In C-elegans sensory ciliary, kinesin-2 motor proteins Kinesin-II complex and OSM-3 homodimer transport IFT trains anterogradely to the ciliary tip. Kinesin-II carries OSM-3 as an inactive passenger from the ciliary base to the middle segment, where kinesin-II dissociates from IFT trains and OSM-3 gets activated and transports IFT trains to the distal segment. Therefore, activation/inactivation of OSM-3 plays an essential role in its ciliary function.

Strengths:

In this study, using mass spectrometry, the authors have shown that the NEKL-3 kinase phosphorylates a serine/threonine patch at the hinge region between coiled coils 1 and 2

of an OSM-3 dimer, referred to as the elbow region in ubiquitous kinesin-1. Phosphomimic mutants of these sites inhibit OSM-3 motility both *in vitro* and *in vivo*, suggesting that this phosphorylation is critical for the autoinhibition of the motor. Conversely, phospho-dead mutants of these sites hyperactivate OSM-3 motility *in vitro* and affect the localization of OSM3 in *C. elegans*. The authors also showed that Alanine to Tyrosine mutation of one of the phosphorylation rescues OS-3 function in live worms.

Weaknesses:

Collectively, this study presents evidence for the physiological role of OSM-3 elbow phosphorylation in its autoregulation, which affects ciliary localization and function of this motor. Overall, the work is well performed, and the results mostly support the conclusions of this manuscript. However, the work will benefit from additional experiments to further support conclusions and rule out alternative explanations, filling some logical gaps with new experimental evidence and in-text clarifications, and improving writing before I can recommend publication.

We appreciate Reviewer #1's comments and suggestions. We have now provided additional evidences and discussions to further support our conclusions and fill the logical gaps. We have also provided alternative explanations to our data and improved writing.

Reviewer #2 (Public review):

Summary:

The regulation of kinesin is fundamental to cellular morphogenesis. Previously, it has been shown that OSM-3, a kinesin required for intraflagellar transport (IFT), is regulated by autoinhibition. However, it remains totally elusive how the autoinhibition of OSM-3 is released. In this study, the authors have shown that NEKL-3 phosphorylates OSM-3 and releases its autoinhibition.

*The authors found NEKL-3 directly phosphorylates OSM-3 (although the method is not described clearly) (Figure 1). The phosphorylated residue is the "elbow" of OSM-3. The authors introduced phospho-dead (PD) and phospho-mimic (PM) mutations by genome editing and found that the OSM-3(PD) protein does not form cilia, and instead, accumulates to the axonal tips. The phenotype is similar to another constitutive active mutant of OSM-3, OSM-3(G444A) (Imanishi et al., 2006; Xie et al., 2024). *osm-3*(PM) has shorter cilia, which resembles with loss of function mutants of *osm-3* (Figure 3). The authors did structural prediction and showed that G444E and PD mutations change the conformation of OSM-3 protein (Figure 3). In the single-molecule assays G444E and PD mutations exhibited increased landing rate (Figure 4). By unbiased genetic screening, the authors identified a suppressor mutant of *osm-3*(PD), in which A489T occurs. The result confirms the importance of this residue. Based on these results, the authors suggest that NEKL-3 induces phosphorylation of the elbow domain and inactivates OSM-3 motor when the motor is synthesized in the cell body. This regulation is essential for proper cilia formation.*

Strengths:

The finding is interesting and gives new insight into how the IFT motor is regulated.

Weaknesses:

The methods section has not presented sufficient information to reproduce this study.

We appreciate that Reviewer #2 is also positive to our study. We have now provided sufficient information in the revised Methods section.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Major Concerns

(1) Why do the authors think that NEKL-3 phosphorylates OSM-3 in the first place? This seems to come out of nowhere and prior evidence indicating that NEKL-3 may be phosphorylating OSM-3 is not even mentioned in the Introduction.

We thank the Reviewer for raising this important point. Our hypothesis that NEKL-3 phosphorylates OSM-3 stems from prior findings in our lab. In a previous study (Yi et al., Traffic, 2018, PMID: 29655266), we identified NEKL-4, a member of the NIMA kinase family, as a suppressor of the OSM-3(G444E) hyperactive mutation. This discovery prompted us to explore the broader role of NIMA kinases in regulating OSM3. Subsequent genetic screens (Xie et al., EMBO J, 2024, PMID: 38806659) revealed that both NEKL-3 and NEKL-4 suppress multiple OSM-3 mutations, further supporting their functional interaction. Given the established role of NIMA kinases in phosphorylation-dependent processes (Fry et al., JCS, 2012, PMID: 23132929; Chivukula et al., Nat. Med., 2020, PMID: 31959991; Thiel, C. et al. Am. J. Hum. Genet. 2011, PMID: 21211617; Smith, L. A. et al., J. Am. Soc. Nephrol., 2006, PMID: 16928806), we hypothesized that NEKL-3/4 may directly phosphorylate OSM-3 to modulate its activity.

To test this hypothesis, we expressed recombinant *C. elegans* NEKL-3 and OSM-3 proteins and conducted in vitro phosphorylation assays. While we were unable to obtain active recombinant NEKL-4 (limitations noted in the revised text), our experiments with NEKL-3 revealed phosphorylation at residues 487-490 (YSTT motif) in OSM-3's tail region, as confirmed by mass spectrometry. These findings are now explicitly contextualized in the Introduction and Results sections of the revised manuscript.

Page #4, Line #11:

"...In our previous study (Yi et al., Traffic, 2018, PMID: 29655266), a genetic screen targeting the OSM-3(G444E) hyperactive mutation identified NEKL-4, a member of the NIMA kinase family, as a suppressor of this phenotype. This finding, combined with reports that NIMA kinases regulate ciliary processes independently of their canonical mitotic roles (Fry et al., JCS, 2012, PMID: 23132929; Chivukula et al., Nat. Med., 2020, PMID: 31959991; Thiel, C. et al. Am. J. Hum. Genet. 2011, PMID: 21211617; Smith, L. A. et al., J. Am. Soc. Nephrol., 2006, PMID: 16928806), prompted us to investigate whether NIMA kinases modulate OSM-3-driven intraflagellar transport. We hypothesized that NEKL-3/4, as paralogs within this family, might directly phosphorylate OSM-3 to regulate its motility..."

Page #4, line #26:

"... To determine whether NIMA kinase family members could directly phosphorylate

OSM-3, we purified prokaryotic recombinant *C. elegans* NEKL-3/NEKL-4 and OSM3 protein in order to perform in vitro phosphorylation assays. We were able to obtain active recombinant NEKL-3 but not NEKL-4. The in vitro phosphorylation assays showed that NEKL-3, directly phosphorylates OSM-3 (Fig. 1A-B, Appendix Table S1). Subsequent mass spectrometric analysis revealed phosphorylation at residues 487-490, which localize to the conserved "YSTT" motif within OSM-3's C-terminal tail region ..."

(2) The authors need to characterize the proteins they expressed and purified for in vitro ATPase and motility assays. Are these proteins monomers or dimers?

For our in vitro ATPase and motility assays, OSM-3 was expressed in *E. coli* BL21(DE3) and purified using established protocols (Xie et al., EMBO J, 2024, PMID: 38806659; Imanishi et al., JCB, 2006, PMID: 17000874). To confirm its oligomeric state, we analyzed recombinant OSM-3 by size-exclusion chromatography coupled with multiangle light scattering (SEC-MALS). As reported in Xie et al. (2024), OSM-3 (~80 kDa monomer) elutes with a molecular weight of 173–193 kDa under physiological buffer conditions, consistent with a homodimeric assembly. These findings confirm that the functional unit used in our assays is the biologically relevant dimer. This characterization has been added to the revised manuscript on Page #35, Line #7.

“...OSM-3 was expressed in *E. coli* BL21(DE3) and purified for in vitro assays using established protocols (REFs). Size-exclusion chromatography coupled with multiangle light scattering (SEC-MALS) (Xie et al., EMBO J., 2024) confirmed that recombinant OSM-3 forms a homodimer (173–193 kDa) under physiological conditions, ensuring its dimeric state remained intact...”

(3) The authors primarily used PD and PM mutations, which affect all four amino acids in the region. This may or may not be physiologically relevant. Figure 5 indicates that T489 is a critical regulatory site. However, this conclusion is undermined by reliance on PD mutations, which affect all four amino acids. Creating PM (T489E) and PD (T489A) mutations based on WT OSM-3 would better reflect physiological relevance. In vitro assays with a single phosphomimic or phosphor-dead mutation at residue 489 are missing at the end of this story. This would better link Figure 5 with the rest of the manuscript.

We thank the reviewer for this constructive critique. Below, we address the concerns and integrate new data to strengthen the link between T489 and autoinhibition:

To probe the regulatory role of T489 phosphorylation, we generated *osm-3*(T489E) (phosphomimetic, PM) and *osm-3*(T489A) (phospho-dead, PD) mutant animals. Strikingly, both mutants formed axonal puncta (Figure S7), recapitulating the hyperactive phenotype of the OSM-3G444E mutant. While the similar puncta formation in PM and PD mutants initially appeared paradoxical, this observation underscores the necessity of dynamic phosphorylation cycling at T489 for proper autoinhibition. Specifically, the PD mutant (T489A) likely disrupts phosphorylation-dependent autoinhibition stabilization, leading to constitutive activation, whereas the PM mutant (T489E) may mimic a “locked” phosphorylated state, preventing dephosphorylation-dependent release of autoinhibition in cilia and trapping OSM-3 in an aggregation-prone conformation. These results highlight T489 as a structural linchpin whose post-translational modification dynamically regulates motor activity. While the precise molecular mechanism—such as how phosphorylation modulates tailmotor domain interactions—remains to be elucidated, our data conclusively demonstrate that perturbing T489 (even in isolation) destabilizes autoinhibition, driving puncta formation and the constitutive activity.

We have integrated the above paragraph in the revised manuscript on page #8, line #27.

(4) There seems to be a disconnect between the MT gliding assays in Figure 4C and single molecule motility assays in Figure 4E. The gliding assays show that all constructs can glide microtubules at near WT speeds. Yet, the motility assays show that WT and PM cannot land or walk on MTs. The authors need to explain why this is the case. Is this because surface immobilization of kinesin from its tail disrupts autoinhibition? Alternatively, the protein preparation may include monomers that cannot be autoinhibited and cannot land and processively walk on surface-immobilized microtubules (because they only have one motor domain) but can glide microtubules when immobilized on the surface from their tail.

The surface immobilization of OSM-3 via its tail domain disrupts autoinhibition, a phenomenon previously observed in other kinesins such as kinesin-1 (Nitzsche et al, *Methods Cell Biol.*, 2010, PMID: 20466139). In our assays, OSM-3 was nonspecifically immobilized on glass surfaces, enabling microtubule gliding by motors whose autoinhibition was relieved through tail anchoring. Critically, the PD and PM mutations reside in the tail region and do not alter the intrinsic properties of the motor head domain. Consequently, once autoinhibition is released via immobilization, the gliding velocities reflect the conserved motor head activity, which is expected to remain comparable across all constructs. While we cannot entirely rule out the presence of monomeric OSM-3 in solution, several lines of evidence argue against this possibility. First, the mutations are located in the elbow region, which is dispensable for motor dimerization. Second, SEC-MALS analysis from prior studies confirms that purified OSM-3 exists predominantly as dimers in solution.

We have discussed these issues in the revised text on page #10, line #18:

“...In our gliding assays, OSM-3PM has an increased gliding speed of $0.69 \pm 0.07 \mu\text{m/s}$ (Fig. 4 C-D), similar to PD mutant. PD and PM mutations are confined to the elbow region, leaving the motor head’s mechanochemical properties intact. Upon tail immobilization—which releases autoinhibition—the gliding speeds reflect motor head activity. Single-molecule assays, however, directly resolve their native regulatory states: PD mutants are constitutively active, whereas PM mutants persist in an autoinhibited state (Fig. 4E-G). Although monomeric OSM-3 could theoretically mediate singlemotor gliding, the previous SEC-MALS data demonstrate that OSM-3 purifies as stable dimers (Xie et al., *EMBO J*, 2024, PMID: 38806659). Thus, dimeric OSM-3 is perhaps the predominant functional species in our assays...”

(5) An alternative explanation for the data is that both PD and PM mutations result in loss-of-function effects, disrupting OSM-3 activity. For instance:

a) In Figure 2C, both mutations cause shorter cilia than the wild type (WT).

b) In Figure 4A, both mutations result in higher ATPase activity than WT.

c) In Figure 4D, both mutations show increased gliding velocity compared to WT. These results suggest the observed effects could stem from loss of function rather than phosphorylation-specific regulation.

Although PD and PM mutations exhibit superficially similar “loss-of-function” phenotypes in certain assays, they mechanistically disrupt motor regulation in distinct ways:

a) Ciliary Length (Figure 2C) PD Mutants: Hyperactivation causes OSM-3-PD to prematurely aggregate into axonal puncta, preventing ciliary entry. Consequently, cilia are built solely by the weaker Kinesin-II motor, which only constructs shorter middle segments.

PM Mutants: OSM-3-PM retains autoinhibition during transport (enabling ciliary entry) but cannot be dephosphorylated in cilia. This blocks activation, leaving OSM-3-PM partially functional and resulting in cilia intermediate in length between WT and PD.

We have discussed this issue in the revised text on page #5, line #30:

“...These findings indicate that OSM-3-PM is in an autoinhibited state capable of ciliary delivery, yet fails to achieve full activation due to defective dephosphorylation. This incomplete activation results in suboptimal motor function and intermediate ciliary length phenotypes (Fig.2 B-C). In contrast, OSM-3-PD exhibits constitutive activation leading to aggregation into axonal puncta, which completely abolishes its ciliary entry capacity (Fig.2 A-B).”

b) ATPase Activity (Figure 4A)

PD Mutants: Fully autoinhibition-released (98.15% of KHC ATPase activity), consistent with constitutive activation.

PM Mutants: Show partial ATPase activity (34.28% of KHC), reflecting imperfect phosphomimicry. While the DDEE substitution introduces negative charges, it fails to fully replicate the steric/kinetic effects of phosphorylated tyrosine (Y486; phenyl ring absent), resulting in incomplete autoinhibition stabilization. Despite this, the residual inhibition is sufficient to phenocopy shorter cilia in vivo.

We have discussed this issue in the revised text on page #7, line#19:

“...The PM mutant’s partial ATPase activity (34.28% of KHC) might arise from imperfect phosphomimicry—while the DDEE substitution introduces negative charges, it lacks the steric bulk of phosphorylated tyrosine (pY487). And this incomplete mimicry allows residual autoinhibition, sufficient to limit ciliary construction in vivo...”

c) Microtubule Gliding Velocity (Figure 4D)

Gliding Assay Limitation: Tail immobilization artificially releases autoinhibition, masking regulatory differences. Thus, all constructs (PD, PM) exhibit similar velocities ($\sim 0.7 \mu\text{m/s}$), reflecting conserved motor head activity.

Single-Molecule Assay (Figure 4E): Directly resolves native autoinhibition states:

PD mutants show robust motility (autoinhibition released).

PM mutants remain largely inactive (autoinhibition retained).

We have discussed this issue in the revised text on page #10, line#18:

“...In our gliding assays, OSM-3PM has an increased gliding speed of $0.69 \pm 0.07 \mu\text{m/s}$ (Fig. 4 C-D), similar to PD mutant. PD and PM mutations are confined to the elbow region, leaving the motor head’s mechanochemical properties intact. Upon tail immobilization—which releases autoinhibition—the gliding speeds reflect motor head activity. Single-molecule assays, however, directly resolve their native regulatory states: PD mutants are constitutively active, whereas PM mutants persist in an autoinhibited state (Fig. 4E-G)...”

Minor Suggestions and Concerns

(1) Lines 60-66: References that support these observations are missing from this section.

We have added the relevant references.

(2) Lines 66-67: I would revise this sentence as "It remains unclear how OSM-3 becomes enriched..."

We have made the changes.

(3) Line 85: The authors should describe how they perform these assays (i.e. recombinantly expressed NEKL-3 and OSM-3, are these *C. elegans* proteins, and which expression system was used...).

We have described them in the main text and methods

Page #4 line #26

“...To determine whether NIMA kinase family members could directly phosphorylate OSM-3, we purified prokaryotic recombinant *C. elegans* NEKL-3/NEKL-4 and OSM-3 protein in order to perform in vitro phosphorylation assays...”

Page #35 line#12

“...Basically, point mutations was introduced in to pET.M.3C OSM-3-eGFP-His6 plasmid for prokaryotic expression. Plasmid transformed *E. coli* (BL21) was cultured at 37°C and induced overnight at 23°C with 0.2 mM IPTG. Cells were lysed in lysis buffer (50 mM NaPO₄ pH8.0, 250 mM NaCl, 20 mM imidazole, 10 mM bME, 0.5 mM ATP, 1 mM MgCl₂, Complete Protease Inhibitor Cocktail (Roche)) and Ni-NTA beads were applied for affinity purification. After incubation, beads were washed with wash buffer (50 mM NaPO₄ pH6.0, 250 mM NaCl, 10 mM bME, 0.1 mM ATP, 1 mM MgCl₂) and eluted with elute buffer (50 mM NaPO₄ pH7.2, 250 mM NaCl, 500 mM imidazole, 10 mM bME, 0.1 mM ATP, 1 mM MgCl₂). Protein concentration was determined by standard Bradford assay. *C. elegans* nekl-3 cDNA was cloned in to pGEX-6P GST vector and expressed in *E. coli* BL21 (DE3) and purified for in vitro phosphorylation assays. Plasmid transformed *E. coli* (BL21) was cultured at 37°C and induced overnight at 18°C with 0.5 mM IPTG. Cells were lysed in lysis buffer (50 mM NaPO₄ pH8.0, 250 mM NaCl, 1 mM DTT, Complete Protease Inhibitor Cocktail (Roche)) and GST beads were applied for affinity purification. After incubation, beads were washed with wash buffer (50 mM NaPO₄ pH6.0, 250 mM NaCl, 1 mM DTT) and eluted with elute buffer (50 mM NaPO₄ pH7.2, 150 mM NaCl, 10 mM GSH, 1 mM DTT). Purified proteins were dialyzed against storage buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl). Protein concentration was determined by standard Bradford assay...”

(4) Line 141: The first sentence of this paragraph lacks motivation. I would start this sentence with "To directly observe the effects of phosphor mutants in the elbow region in microtubule binding and motility of OSM-3, we..."

We have made the change.

(5) Figure 1B: The mass spectrometry data in Figure 1B lacks adequate explanation. The Methods section should detail the experimental protocol, data interpretation, and any databases used. Additionally, the manuscript should list all identified phosphorylation sites on OSM-3 to provide context, including whether Y487_T490 is the major site.

We have provided the detailed experimental protocol, data interpretation, and databases used in methods. We have provided all identified sites as Appendix table S1.

(6) Figure 1C: Is it possible to model the effect of PM and PD mutations using AlphaFold? The authors should also show PAE or pLDDT scores of their model.

AlphaFold cannot well model the effect of mutants, but we conducted the Rosetta relax to capture their possible conformational changes, as shown in the revised Figure 3. We have provided PAE and pLDDT as a new figure, Figure S2.

(7) Figure 2D: The unit for speed should use a lowercase "s" for seconds.

We have fixed it.

(8) Figure 3: I am not sure whether this figure stands for a main text figure on its own, as it is only a Rosetta prediction and is not supported by any experimental data. In addition, it remains unclear what the labels on the x-axis mean.

We have updated the figure and explain the labels on the x-axis in Figure S4 to make it more reader-friendly.

(9) *Figure 4: NEKL-3-treated OSM-1 should be included as a positive control in the in vitro experiments.*

We suspect that the Reviewer asked for NEKL-3-treated OSM-3.

In our other study which has just been accepted by the Journal of Cell Biology, NEKL3-treated OSM-3 significantly reduced the affinity between OSM-3 motor and microtubules and showed very low ATPase activity. We have cited and discussed this in the revised text on page #10, line #28:

“...As demonstrated in our recent study (Huang et al., JCB, 2025, In press, attached), phosphorylation of OSM-3 by NEKL-3 at two distinct regions—Ser96 and the conserved “elbow” motif—differentially regulates its activity and localization. Phosphorylation at Ser96 reduces OSM-3’s ATPase activity and alters its ciliary distribution from the distal segment to a uniform localization, while elbow phosphorylation induces autoinhibition, retaining OSM-3 in the cell body. Strikingly, in vitro phosphorylation of OSM-3 by NEKL-3 significantly reduces its microtubulebinding affinity, likely arising from combined modifications at both sites. We propose a model wherein elbow phosphorylation ensures anterograde ciliary transport, while Ser96 phosphorylation fine-tunes distal segment targeting. This multistep regulation may involve distinct phosphatases to reverse phosphorylation at specific sites, a hypothesis warranting further investigation....”

(10) *Figure 4C, D, and F: The unit of velocity is wrong. The authors should use the same units they used in the table shown in Figure 4B.*

We have fixed these errors

(11) *Figure 4F: The velocity of PD is a lot lower than G444E. Therefore, it would be more appropriate to refer to PD as partially active, rather than hyperactive.*

We have made the change.

(12) *Figure 5: There is too much genetics jargon on this figure (EMF, F2, 100%Dyf,...). How are the alleles numbered? Is it OK to refer to them as Alleles 1 and 2 for simplicity?*

According to the established *C. elegans* allele nomenclature, each worm allele has a unique number named after the lab code for identification. We have simplified the labels and updated the figure to make it more reader-friendly.

(13) *Figure 5E: A plot would be more reader-friendly than a table. Additionally, the legend for Fig. 5E mistakenly refers to it as “D.”*

We have changed the table to a plot and fixed the mistakes. We thank the Reviewer for pointing them out.

Reviewer #2 (Recommendations for the authors):

(1) *The model appears as if NEKL-3 induces dephosphorylation of OSM-3 (Figure 6). This is not consistent with the conclusions described in the Discussion and is confusing.*

We have updated the model figure and fixed the error.

(2) *It should be described why the authors hypothesized NEKL-3 phosphorylates OSM3. Was there genetic evidence? Did the authors screened cilia-related kinases? or Did the authors identify it incidentally? Providing this information would help readers to understand the context of the research.*

We appreciate both Reviewers for pointing out this issue.

Our hypothesis that NEKL-3 phosphorylates OSM-3 stems from prior findings in our lab. In a previous study (Yi et al., Traffic, 2018, PMID: 29655266), we identified NEKL-4, a member of the NIMA kinase family, as a suppressor of the OSM-3(G444E) hyperactive mutation. This discovery prompted us to explore the broader role of NIMA kinases in regulating OSM-3. Subsequent genetic screens (Xie et al., EMBO J, 2024, PMID: 38806659) revealed that both NEKL-3 and NEKL-4 suppress multiple OSM-3 mutations, further supporting their functional interaction. Given the established role of NIMA kinases in phosphorylation-dependent processes (Fry et al., JCS, 2012, PMID: 23132929; Chivukula et al., Nat. Med., 2020, PMID: 31959991; Thiel, C. et al. Am. J. Hum. Genet. 2011, PMID: 21211617; Smith, L. A. et al., J. Am. Soc. Nephrol., 2006, PMID: 16928806), we hypothesized that NEKL-3/4 may directly phosphorylate OSM3 to modulate its activity.

To test this hypothesis, we expressed recombinant *C. elegans* NEKL-3 and OSM-3 proteins and conducted in vitro phosphorylation assays. While we were unable to obtain active recombinant NEKL-4 (limitations noted in the revised text), our experiments with NEKL-3 revealed phosphorylation at residues 487-490 (YSTT motif) in OSM-3's tail region, as confirmed by mass spectrometry. These findings are now explicitly contextualized in the Introduction and Results sections of the revised manuscript.

Page #4, Line #11:

"... In our previous study (Yi et al., Traffic, 2018, PMID: 29655266), a genetic screen targeting the OSM-3(G444E) hyperactive mutation identified NEKL-4, a member of the NIMA kinase family, as a suppressor of this phenotype. This finding, combined with reports that NIMA kinases regulate ciliary processes independently of their canonical mitotic roles (Fry et al., JCS, 2012, PMID: 23132929; Chivukula et al., Nat. Med., 2020, PMID: 31959991; Thiel, C. et al. Am. J. Hum. Genet. 2011, PMID: 21211617; Smith, L. A. et al., J. Am. Soc. Nephrol., 2006, PMID: 16928806), prompted us to investigate whether NIMA kinases modulate OSM-3-driven intraflagellar transport. We hypothesized that NEKL-3/4, as paralogs within this family, might directly phosphorylate OSM-3 to regulate its motility..."

Page #4, line #26:

"... To determine whether NIMA kinase family members could directly phosphorylate OSM-3, we purified prokaryotic recombinant *C. elegans* NEKL-3/NEKL-4 and OSM3 protein in order to perform in vitro phosphorylation assays. We were able to obtain active recombinant NEKL-3 but not NEKL-4. The in vitro phosphorylation assays showed that NEKL-3, directly phosphorylates OSM-3 (Fig. 1A-B, Appendix Table S1). Subsequent mass spectrometric analysis revealed phosphorylation at residues 487-490, which localize to the conserved "YSTT" motif within OSM-3's C-terminal tail region..."

(3) *It is curious the authors have not addressed the cilia phenotype and the localization of OSM-3 in nekl-3 mutant. Regardless of whether these observations agrees with the proposed mechanisms, it is essential for the authors to show and discuss the cilia phenotype and OSM-3 localization in nekl-3 mutants.*

We thank the Reviewer for highlighting this critical point. Indeed, *nekl-3* null mutants are inviable due to essential mitotic roles (Barstead et al., 2012, PMID: 23173093), precluding

direct analysis of ciliary phenotypes. To bypass this limitation, we recently generated *nekl-3* conditional knockouts (cKOs) in ciliated neurons (Huang et al., JCB, 2025 in press, attached). In these mutants, OSM-3—which is normally enriched in the ciliary distal segment—becomes uniformly distributed along the cilium. This redistribution correlates with premature activation of OSM-3-driven anterograde motility in the ciliary middle region, consistent with our proposed model where NEKL3 phosphorylation suppresses OSM-3 activity. We have now integrated this result and discussion into the revised manuscript, reinforcing the physiological relevance of NEKL-3-mediated regulation in ciliary transport.

Page #6 line #10

“... While *nekl-3* null mutants are inviable due to essential mitotic roles (Barstead et al., 2012, PMID: 23173093), conditional knockout (cKO) of *nekl-3* in ciliated neurons (Huang et al., JCB, 2025 in press, attached) revealed its critical role in regulating OSM3 dynamics. In *nekl-3* cKO animals, OSM-3—normally enriched in the ciliary distal segment—redistributed uniformly along the cilium, concomitant with premature activation of anterograde motility in the middle ciliary region. This phenotype aligns with our model wherein NEKL-3 phosphorylation suppresses OSM-3 activity, ensuring spatiotemporal regulation of IFT....”

(4) The methods section lacks some information, which is critical to reproducing this study.

We have now provided detailed information in the methods section in the revised manuscript.

(a) It is not described how the authors determined phosphorylation of OSM-3 by NEKL-3. In methods, nothing is described about the assay.

We performed in vitro phosphorylation assays using recombinant OSM-3 and NEKL3 purified from bacteria. We then used LC-MS/MS for identification of phosphorylation sites. We have now updated the methods section to include all the information.

Page #4 line #26

“... To determine whether NIMA kinase family members could directly phosphorylate OSM-3, we purified prokaryotic recombinant *C. elegans* NEKL-3/NEKL-4 and OSM3 protein in order to perform in vitro phosphorylation assays. We were able to obtain active recombinant NEKL-3 but not NEKL-4. The in vitro phosphorylation assays showed that NEKL-3, directly phosphorylates OSM-3 (Fig. 1A-B, Appendix Table S1). Subsequent mass spectrometric analysis revealed phosphorylation at residues 487-490, which localize to the conserved “YSTT” motif within OSM-3’s C-terminal tail region...”

Page #36, line #19

“In vitro phosphorylation assay 20 μ M purified OSM-3 was incubated with 1 μ M GST-NEKL-3 at 30 °C in 100 μ L reaction buffer (50 mM Tris-HCl pH 8.0, 10 mM MgCl₂, 150 mM NaCl, and 2 mM ATP) for 30 min. The reaction was terminated by boiling for 5 min with an SDS-sample buffer.

Mass spectrometry

Following NEKL-3 treatment, OSM-3 proteins were resolved by SDS-PAGE and visualized with Coomassie Brilliant Blue staining. Protein bands corresponding to OSM-3 were excised and subjected to digestion using the following protocol: reduction with 5 mM TCEP at 56°C for 30 min; alkylation with 10 mM iodoacetamide in darkness for 45 min at room temperature, and tryptic digestion at 37°C overnight with a 1:20 enzyme-to-protein ratio. The resulting peptides were subjected to mass spectrometry analysis. Briefly, the peptides were analyzed using an

UltiMate 3000 RSLCnano system coupled to an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific). We applied an in-house proteome discovery searching algorithm to search the MS/MS data against the *C. elegans* database. Phosphorylation sites were determined using PhosphoRS algorithm with manual validation of MS/MS spectra.”

(b) The method of structural prediction by AlphaFold2 and LocalColabFold needs clarification. In general, the prediction gives several candidates. How did the authors choose one of these candidates?

We generated five candidate models and all of them showed similar conformation. We thus chose the model with the highest confidence. We have provided PAE and pLDDT as additional data in Figure S2 and discussed them in the revised text on, Page #4, line #32:

“...To gain structural insights from this motif, we employed LocalColabFold based on AlphaFold2 to predict the dimeric structure of OSM-3 (Evans et al., 2022; Jumper et al., 2021; Mirdita et al., 2022). The highest-confidence model was selected for further analysis (Fig. 1C, Fig. S2)...”

(c) The methods to predict conformational changes by introducing various point mutations are interesting (Figure 3). However, the methods require more detailed descriptions. In the current form, the manuscript only lists the tools used. The pipelines and parameters need to be described. This information is important because AlphaFoldbased predictions often give folded conformations because the training data are mainly composed of folded proteins. It is surprising that the methods applied here give open conformations induced by point mutations.

We have described the pipelines in the revised Methods section on page#34, line#25:

“...OSM-3 model was predicted using LocalColabFold (Evans et al., 2022; Jumper et al., 2021; Mirdita et al., 2022). Mutated proteins were designed by Pymol 2.6, choosing the rotamer of the mutated residues in G444E, PM and PD models with the least clash as the initial conformation. To predict mutation-induced conformational changes, the initial models were subjected to Pyrosetta (Chaudhury et al., 2010). The energies of pre-relaxed models were evaluated with Rosetta Energy Function 2015 (Alford et al., 2017), and then the relax procedure were applied to the models with default parameters to obtain the relaxed models visualized by Pymol to minimize the energy of these models. In detail, to obtain the relaxed models visualized by Pymol and minimize the energy of these models, the classic relax mover was used in the procedure mentioned above with default settings. The relax script has been uploaded to Github: https://github.com/young55775/RosettaRelax_for_OSM3...”

(5) The authors have purified proteins. Do they show different properties in gel filtration that are consistent with the structural prediction? It is anticipated that open-form mutants are eluted from earlier than closed forms.

We thank the reviewer for this insightful suggestion. Indeed, our recent study supported that the open-form of the active OSM-3 G444E mutation were eluted earlier than the wild-type closed form (Xie et al., EMBO J., 2024). While the current study did not perform gel filtration chromatography (SEC) to directly compare the hydrodynamic properties of the OSM-3 mutants, our functional assays provide robust evidence for conformational changes predicted by structural modeling. For example: ATPase activity assays revealed that the open-state mutants (e.g., G444E and PD mutants) exhibited significantly enhanced enzymatic activity (Figure 4A), consistent with structural predictions of an active, destabilized autoinhibitory interface (Figure 3A). These functional readouts collectively validate the predicted structural states. While SEC could further corroborate these findings by distinguishing compact (closed) versus extended (open) conformations, we prioritized assays

that directly link structural predictions to in vitro enzymatic activity and in vivo ciliary transport dynamics. Future studies incorporating SEC or cryo-EM will provide additional biophysical validation of these states.

We have revised the text in the manuscript (Page #7, Lines #22):

“...Notably, the open-state OSM-3 mutants (e.g., G444E) displayed elevated ATPase activity, consistent with structural predictions of autoinhibition release (Fig. 3A, Fig. 4A) (Xie et al., 2024). While hydrodynamic profiling (e.g., SEC) could further resolve conformational states, our functional assays directly connect predicted structural changes to altered biochemical and cellular activity...”

Minor point

(1) Line 85 "MIMA kinase family" should be "NIMA kinase family".

We have corrected the typo and appreciate that the Reviewer for pointing it out.

(2) M.S. and D.S. need to be defined in Figure 2D.

We have updated the figures.

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